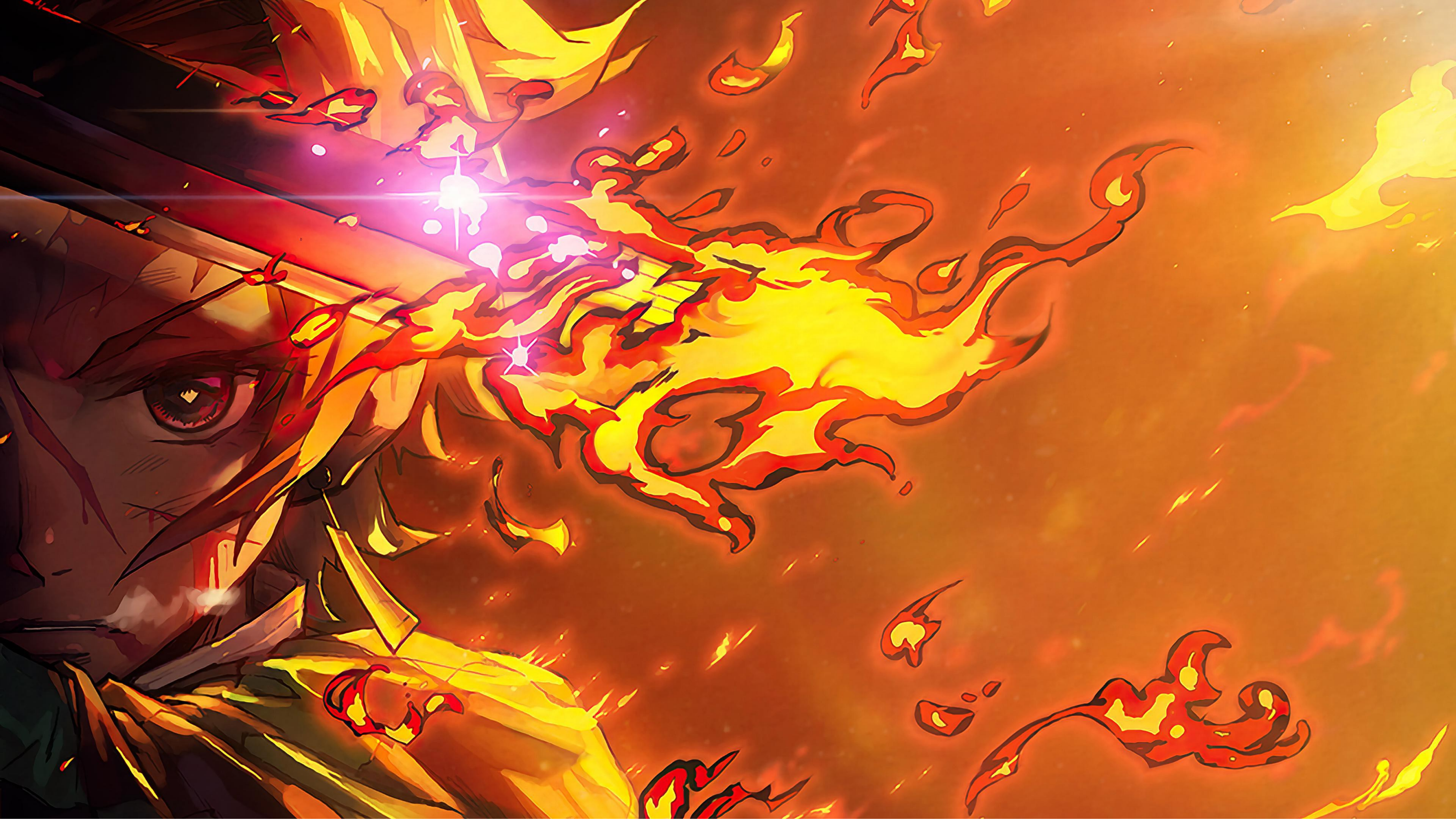

Regression Analysis Note

Author: Yukina

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Chapter 1

Linear Models & OLS

§ 1.1 Linear Models and OLS

1.1.1 Model Specification

Definition 1.1 Let $(Y_i; X_{i1}, \cdots, X_{ip})$ $(1 \leqslant i \leqslant n)$ be n observations. The linear regression model is:

$$Y_i = \beta_0 + \beta_1 X_{i1} + \cdots + \beta_p X_{ip} + \varepsilon_i, \quad i = 1, \cdots, n.$$

In matrix form:

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon},$$

where

$$\mathbf{Y} = \begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{pmatrix}, \quad \mathbf{X} = \begin{pmatrix} 1 & X_{11} & \cdots & X_{1p} \\ 1 & X_{21} & \cdots & X_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & X_{n1} & \cdots & X_{np} \end{pmatrix}, \quad \boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{pmatrix}, \quad \boldsymbol{\varepsilon} = \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{pmatrix}.$$

Assumption: $\mathbf{X}$ has full column rank, i.e., $\text{rank}(\mathbf{X}) = p + 1 < n$.

Definition 1.2 (Gauss-Markov Assumptions) The error vector $\boldsymbol{\varepsilon}$ satisfies:

1. $\mathbb{E}[\boldsymbol{\varepsilon}] = \mathbf{0}$ (Unbiasedness).
2. $\text{Var}(\boldsymbol{\varepsilon}) = \mathbb{E}[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}^\top] = \sigma^2 \mathbf{I}_n$ (Homoscedasticity and Uncorrelatedness).

1.1.2 Estimation of β and σ^2

OLS Estimation

Theorem 1.3 (OLS Solution) The Ordinary Least Squares (OLS) estimator $\hat{\boldsymbol{\beta}}$ minimizes the Error Sum of Squares:

$$\text{SSE}(\boldsymbol{\beta}) = (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})^\top (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}).$$

It is uniquely given by:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}.$$

Proof. The objective function is strictly convex. Differentiating $\mathrm{SSE}(\boldsymbol{\beta})$ with respect to $\boldsymbol{\beta}$:

$$\frac{\partial \mathrm{SSE}(\boldsymbol{\beta})}{\partial \boldsymbol{\beta}} = \frac{\partial}{\partial \boldsymbol{\beta}} \left(\mathbf{Y}^\top \mathbf{Y} - 2\mathbf{Y}^\top \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{X}\boldsymbol{\beta} \right) = -2\mathbf{X}^\top \mathbf{Y} + 2\mathbf{X}^\top \mathbf{X}\boldsymbol{\beta}.$$

Setting the gradient to $\mathbf{0}$ yields the **Normal Equations**: $\mathbf{X}^\top \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}^\top \mathbf{Y}$. Since $\mathbf{X}$ is full rank, $\mathbf{X}^\top \mathbf{X}$ is invertible, yielding the unique solution. □

Geometric Interpretation and Residuals

Definition 1.4 (Hat Matrix) The Hat matrix (Projection Matrix) is defined as :

$$\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top.$$

Remark 1.5

$$\mathbf{H}^2 = \mathbf{H} \quad \text{and} \quad \mathbf{H}^\top = \mathbf{H}$$

The hat matrix is symmetric and idempotent.

Definition 1.6 Fitted Values and Residuals:

- Fitted values: $\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{H}\mathbf{Y}$.
- Residuals: $\hat{\boldsymbol{\varepsilon}} = \mathbf{Y} - \hat{\mathbf{Y}} = (\mathbf{I} - \mathbf{H})\mathbf{Y} = (\mathbf{I} - \mathbf{H})\boldsymbol{\varepsilon}$.
- Residual Sum of Squares (SSE):

$$\mathrm{SSE} = \hat{\boldsymbol{\varepsilon}}^\top \hat{\boldsymbol{\varepsilon}} = \mathbf{Y}^\top (\mathbf{I} - \mathbf{H})\mathbf{Y}.$$

Proposition 1.7 Properties of Residuals:

1. $\mathbf{X}^\top \hat{\varepsilon} = \mathbf{0}$ (Orthogonality to predictors).
2. $\hat{\mathbf{Y}}^\top \hat{\varepsilon} = 0$ (Orthogonality to fitted values).
3. $\sum_{i=1}^n \hat{\varepsilon}_i = 0$ (if intercept is included).

Estimation of Error Variance

Theorem 1.8 An unbiased estimator for σ^2 is:

$$\hat{\sigma}^2 = \frac{\mathrm{SSE}}{n - p - 1}.$$

Proof. Using the trace trick :

$$\mathbb{E}[z^\top \mathbf{A} z] = \operatorname{tr}(\mathbf{A}\operatorname{Cov}(z)) + \mathbb{E}[z]^\top \mathbf{A}\mathbb{E}[z]$$

F or ε:

$$\begin{aligned}\mathbb{E}[\mathrm{SSE}] &= \mathbb{E}[\boldsymbol{\varepsilon}^\top(\mathbf{I}-\mathbf{H})\boldsymbol{\varepsilon}] \\ &= \operatorname{tr}((\mathbf{I}-\mathbf{H})\sigma^2\mathbf{I}) + \mathbf{0} \\ &= \sigma^2(\operatorname{tr}(\mathbf{I}_n) - \operatorname{tr}(\mathbf{H})).\end{aligned}$$

Since $\operatorname{tr}(\mathbf{H}) = \operatorname{tr}(\mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top) = \operatorname{tr}((\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{X}) = \operatorname{tr}(\mathbf{I}_{p+1}) = p+1$, we have $\mathbb{E}[\mathrm{SSE}] = \sigma^2(n-p-1)$. □

1.1.3 Gauss-Markov Theorem

Theorem 1.9 (Gauss-Markov) Under the Gauss-Markov assumptions, $\hat{\boldsymbol{\beta}}$ is the Best Linear Unbiased Estimator (BLUE) of $\boldsymbol{\beta}$.

1. $\mathbb{E}[\hat{\boldsymbol{\beta}}] = \boldsymbol{\beta}$ (Unbiased).
2. $\operatorname{Var}(\hat{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1}$.
3. For any other linear unbiased estimator $\tilde{\boldsymbol{\beta}}$, $\operatorname{Var}(\tilde{\boldsymbol{\beta}}) - \operatorname{Var}(\hat{\boldsymbol{\beta}})$ is positive semi-definite.

Proof. (1)&(2): We have:

$$\begin{aligned}\mathbb{E}[\hat{\boldsymbol{\beta}}] &= (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbb{E}[\mathbf{X}\boldsymbol{\beta}+\boldsymbol{\varepsilon}] = \boldsymbol{\beta}. \\ \operatorname{Var}(\hat{\boldsymbol{\beta}}) &= (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top(\sigma^2\mathbf{I})\mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1} = \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1}\end{aligned}$$

(3): Let $\tilde{\boldsymbol{\beta}} = \mathbf{CY}$ be another linear unbiased estimator. Since it's unbiased:

$$\mathbb{E}[\tilde{\boldsymbol{\beta}}] = \mathbf{C}\mathbb{E}[\mathbf{Y}] = \mathbf{CX}\boldsymbol{\beta} = \boldsymbol{\beta} \implies \mathbf{CX} = \mathbf{I}.$$

Let $\mathbf{D} = \mathbf{C} - (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top$. Note that:

$$\mathbf{DX} = \mathbf{CX} - (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{X} = \mathbf{I} - \mathbf{I} = \mathbf{0}$$

Then:

$$\begin{aligned}\operatorname{Var}(\tilde{\boldsymbol{\beta}}) = \sigma^2\mathbf{CC}^\top &= \sigma^2\left((\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top + \mathbf{D}\right)\left((\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top + \mathbf{D}\right)^\top \\ &= \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1} + \sigma^2\mathbf{DD}^\top \quad (\text{since } \mathbf{DX} = \mathbf{0}) \\ &= \operatorname{Var}(\hat{\boldsymbol{\beta}}) + \sigma^2\mathbf{DD}^\top \geqslant \operatorname{Var}(\hat{\boldsymbol{\beta}}).\end{aligned}$$

□

1.1.4 Distributions under Normality Assumption

Theorem 1.10 Assume $\boldsymbol{\varepsilon} \sim N(\mathbf{0}, \sigma^2\mathbf{I})$,then:

1. $\hat{\boldsymbol{\beta}} \sim N(\boldsymbol{\beta}, \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1})$, $\hat{\boldsymbol{\varepsilon}} \sim N(\mathbf{0}, \sigma^2(\mathbf{I}-\mathbf{H}))$.
2. $\frac{\text{SSE}}{\sigma^2} \sim \chi^2(n-p-1)$.
3. $\hat{\boldsymbol{\beta}}$ and $\hat{\sigma}^2$ are statistically independent.

Proof. **(2):**Since $(\mathbf{I}-\mathbf{H})$ is a symmetric idempotent matrix, we have:

$$\text{rank}(\mathbf{I}-\mathbf{H}) = \text{tr}(\mathbf{I}-\mathbf{H}) = n-p-1.$$

Therefore, there exists an $n \times n$ orthogonal matrix $\mathbf{P}$ such that:

$$\mathbf{P}^T(\mathbf{I}-\mathbf{H})\mathbf{P} = \begin{pmatrix} \mathbf{I}_{n-p-1} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{pmatrix}.$$

Let:

$$\boldsymbol{\eta} = (\eta_1, \eta_2, \cdots, \eta_n)^T = \mathbf{P}^T\boldsymbol{\varepsilon},$$

then $E(\boldsymbol{\eta}) = \mathbf{0}$, $\text{Var}(\boldsymbol{\eta}) = \sigma^2\mathbf{P}^T\mathbf{P} = \sigma^2\mathbf{I}$, so $\boldsymbol{\eta} \sim N(\mathbf{0}, \sigma^2\mathbf{I})$, and:

$$\frac{1}{\sigma^2}SSE = \frac{1}{\sigma^2}\boldsymbol{\eta}^T\mathbf{P}^T(\mathbf{I}-\mathbf{H})\mathbf{P}\boldsymbol{\eta} = \frac{1}{\sigma^2}(\eta_1^2 + \eta_2^2 + \cdots + \eta_{n-p-1}^2).$$

Since $\eta_1, \eta_2, \cdots, \eta_{n-p-1}$ are independent $N(0, \sigma^2)$ random variables, we have:

$$\frac{SSE}{\sigma^2} \sim \chi^2_{n-p-1}.$$

(3): Note that $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\boldsymbol{\varepsilon}$. Consider the joint linear transformation of $\boldsymbol{\varepsilon}$:

$$\begin{pmatrix} \hat{\boldsymbol{\beta}} - \boldsymbol{\beta} \\ (\mathbf{I}-\mathbf{H})\boldsymbol{\varepsilon} \end{pmatrix} = \begin{pmatrix} (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top \\ \mathbf{I}-\mathbf{H} \end{pmatrix}\boldsymbol{\varepsilon}.$$

The covariance between these two blocks is:

$$\text{Cov}(\hat{\boldsymbol{\beta}}, \hat{\boldsymbol{\varepsilon}}) = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top(\sigma^2\mathbf{I})(\mathbf{I}-\mathbf{H})^\top = \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1}(\mathbf{X}^\top - \mathbf{X}^\top) = \mathbf{0}.$$

Since they are jointly normal and uncorrelated, $\hat{\boldsymbol{\beta}}$ and residuals $\hat{\boldsymbol{\varepsilon}}$ (and thus SSE) are independent. □

1.1.5 Constrained Linear Regression

Consider the model $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ subject to linear constraints $\mathbf{A}\boldsymbol{\beta} = \mathbf{b}$, where $\mathbf{A} \in \mathbb{R}^{m\times(p+1)}$ has full row rank m.

Theorem 1.11 (Constrained Estimator) The constrained estimator $\hat{\boldsymbol{\beta}}_c$ is:

$$\hat{\boldsymbol{\beta}}_c = \hat{\boldsymbol{\beta}} - (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{A}^\top \left[\mathbf{A}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{A}^\top\right]^{-1} (\mathbf{A}\hat{\boldsymbol{\beta}} - \mathbf{b}).$$

This estimator satisfies:

1. $\mathbf{A}\hat{\boldsymbol{\beta}}_c = \mathbf{b}$ (constraint satisfaction)
2. For all $\boldsymbol{\beta}$ satisfying $\mathbf{A}\boldsymbol{\beta} = \mathbf{b}$, we have $\|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2 \geqslant \|\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}_c\|^2$ (optimality)

Proof. The Lagrangian is:

$$\mathcal{L}(\boldsymbol{\beta}, \boldsymbol{\lambda}) = (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})^T(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}) + 2\boldsymbol{\lambda}^T(\mathbf{A}\boldsymbol{\beta} - \mathbf{b}),$$

Take partial derivatives:

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial \boldsymbol{\beta}} &= -2\mathbf{X}^T\mathbf{Y} + 2\mathbf{X}^T\mathbf{X}\boldsymbol{\beta} + 2\mathbf{A}^T\boldsymbol{\lambda} = \mathbf{0},\\ \frac{\partial \mathcal{L}}{\partial \boldsymbol{\lambda}} &= 2(\mathbf{A}\boldsymbol{\beta} - \mathbf{b}) = \mathbf{0}.\end{aligned}$$

This gives the system:

$$\mathbf{X}^T\mathbf{X}\boldsymbol{\beta} + \mathbf{A}^T\boldsymbol{\lambda} = \mathbf{X}^T\mathbf{Y}, \tag{1.1}$$
$$\mathbf{A}\boldsymbol{\beta} = \mathbf{b}. \tag{1.2}$$

From equation (1.1):

$$\hat{\boldsymbol{\beta}}_c = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{Y} - (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{A}^T\hat{\boldsymbol{\lambda}} = \hat{\boldsymbol{\beta}} - (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{A}^T\hat{\boldsymbol{\lambda}}.$$

Substitute into equation (1.2):

$$\mathbf{A}\hat{\boldsymbol{\beta}} - \mathbf{A}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{A}^T\hat{\boldsymbol{\lambda}} = \mathbf{b}.$$

Solve for $\hat{\boldsymbol{\lambda}}$:

$$\hat{\boldsymbol{\lambda}} = (\mathbf{A}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{A}^T)^{-1}(\mathbf{A}\hat{\boldsymbol{\beta}} - \mathbf{b}).$$

Substitute back to obtain the constrained estimator:

$$\hat{\boldsymbol{\beta}}_c = \hat{\boldsymbol{\beta}} - (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{A}^T(\mathbf{A}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{A}^T)^{-1}(\mathbf{A}\hat{\boldsymbol{\beta}} - \mathbf{b}).$$

(i) Direct computation shows $\mathbf{A}\hat{\boldsymbol{\beta}}_c = \mathbf{b}$.
(ii) For any $\boldsymbol{\beta}$ satisfying $\mathbf{A}\boldsymbol{\beta} = \mathbf{b}$, consider the decomposition:

$$\begin{aligned}\|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2 &= \|\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}_c + \mathbf{X}(\hat{\boldsymbol{\beta}}_c - \boldsymbol{\beta})\|^2\\ &= \|\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}_c\|^2 + \|\mathbf{X}(\hat{\boldsymbol{\beta}}_c - \boldsymbol{\beta})\|^2 + 2(\hat{\boldsymbol{\beta}}_c - \boldsymbol{\beta})^T\mathbf{X}^T(\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}_c).\end{aligned}$$

The cross term vanishes because:

$$\begin{aligned}\mathbf{X}^T(\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}_c) &= \mathbf{X}^T\mathbf{Y} - \mathbf{X}^T\mathbf{X}\hat{\boldsymbol{\beta}}_c \\ &= \mathbf{A}^T\hat{\boldsymbol{\lambda}} \quad \text{(from equation 1.1)},\end{aligned}$$

and

$$(\hat{\boldsymbol{\beta}}_c - \boldsymbol{\beta})^T\mathbf{A}^T\hat{\boldsymbol{\lambda}} = (\mathbf{A}\hat{\boldsymbol{\beta}}_c - \mathbf{A}\boldsymbol{\beta})^T\hat{\boldsymbol{\lambda}} = (\mathbf{b} - \mathbf{b})^T\hat{\boldsymbol{\lambda}} = 0.$$

Therefore, $\|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2 \geqslant \|\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}_c\|^2$ for all feasible $\boldsymbol{\beta}$. □

Remark 1.12 (Geometric Interpretation) The constrained estimator can be viewed as the projection of the unconstrained estimator onto the constraint subspace $\{\boldsymbol{\beta} : \mathbf{A}\boldsymbol{\beta} = \mathbf{b}\}$.

Theorem 1.13 (Properties of $\hat{\beta}_c$) Under normality:

1. $\hat{\boldsymbol{\beta}}_c \sim N(\boldsymbol{\beta}, \sigma^2\mathbf{V}_c)$, where $\mathbf{V}_c \leqslant (\mathbf{X}^\top\mathbf{X})^{-1}$ (variance reduction).

2. $\mathrm{SSE}_c - \mathrm{SSE} = (\mathbf{A}\hat{\boldsymbol{\beta}} - \mathbf{b})^\top[\mathbf{A}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{A}^\top]^{-1}(\mathbf{A}\hat{\boldsymbol{\beta}} - \mathbf{b})$.

3. $(\mathrm{SSE}_c - \mathrm{SSE})/\sigma^2 \sim \chi^2(m)$ under H_0.

§ 1.2 Hypothesis Testing

In this section, we assume the normal linear model:

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim N(\mathbf{0}, \sigma^2\mathbf{I}_n).$$

The design matrix $\mathbf{X}$ is $n \times (p+1)$ with rank $p+1$.

1.2.1 Decomposition of Variance

The variation in the response $\mathbf{Y}$ can be decomposed into the variation explained by the model and the unexplained residual variation.

Definition 1.14 (Sum of Squares) Let $\bar{Y}$ be the sample mean of Y.

1. **Total Sum of Squares (SST):** Measures total variability.

$$\mathrm{SST} = \sum_{i=1}^{n}(Y_i - \bar{Y})^2 = \mathbf{Y}^\top\left(\mathbf{I} - \frac{1}{n}\mathbf{J}\right)\mathbf{Y},$$

where $\mathbf{J} = \mathbf{1}\mathbf{1}^\top$ is the matrix of ones.

2. **Regression Sum of Squares (SSR):** Measures variability explained by the model.

$$\mathrm{SSR} = \sum_{i=1}^{n}(\hat{Y}_i - \bar{Y})^2 = \mathbf{Y}^\top\left(\mathbf{H} - \frac{1}{n}\mathbf{J}\right)\mathbf{Y}.$$

3. **Error Sum of Squares (SSE):** Measures unexplained variability.

$$\mathrm{SSE} = \sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2 = \hat{\boldsymbol{\varepsilon}}^\top \hat{\boldsymbol{\varepsilon}} = \mathbf{Y}^\top(\mathbf{I} - \mathbf{H})\mathbf{Y}.$$

Remark 1.15

$$\mathrm{SST} = \mathrm{SSR} + \mathrm{SSE}.$$

1.2.2 General Linear Hypothesis

We seek to test linear restrictions on the parameters $\boldsymbol{\beta}$. This framework unifies t-tests, F-tests for regression, and ANOVA.

Definition 1.16 (General Linear Hypothesis) Consider the null hypothesis H_0 against the alternative H_1:

$$H_0 : \mathbf{A}\boldsymbol{\beta} = \mathbf{b} \quad \text{vs} \quad H_1 : \mathbf{A}\boldsymbol{\beta} \neq \mathbf{b},$$

where $\mathbf{A}$ is a known $m \times (p+1)$ matrix with full row rank m ($m \leqslant p+1$), and $\mathbf{b}$ is a known $m \times 1$ vector.

Theorem 1.17 (The General F-Statistic) Let SSE_F be the error sum of squares of the full (unconstrained) model, and SSE_R be the error sum of squares of the reduced (constrained under H_0) model. Under H_0, the statistic

$$F = \frac{(\mathrm{SSE}_R - \mathrm{SSE}_F)/m}{\mathrm{SSE}_F/(n-p-1)}$$

follows an F-distribution $F_{m,n-p-1}$.

1.2.3 Specific Hypothesis Tests

We apply Theorem 1.17 to common scenarios.

Test for Global Significance

Does the model have any explanatory power?

$$H_0 : \beta_1 = \beta_2 = \cdots = \beta_p = 0 \quad \text{(excluding intercept } \beta_0\text{)}.$$

Here $\mathbf{A} = [\mathbf{0}, \mathbf{I}_p]$ and $\mathbf{b} = \mathbf{0}$.

- **Reduced Model:** $Y_i = \beta_0 + \varepsilon_i$. The estimate is $\hat{\beta}_0 = \bar{Y}$, so $\mathrm{SSE}_R = \mathrm{SST}$.
- **Full Model:** $\mathrm{SSE}_F = \mathrm{SSE}$.
- **Numerator:** $\mathrm{SSE}_R - \mathrm{SSE}_F = \mathrm{SST} - \mathrm{SSE} = \mathrm{SSR}$.

The F-statistic becomes:

$$F = \frac{\mathrm{SSR}/p}{\mathrm{SSE}/(n-p-1)}.$$

Test for a Subset of Coefficients (Partial F-Test)

Is a subset of predictors significant given the others?

$$H_0 : \beta_{k+1} = \cdots = \beta_p = 0.$$

Partition predictors as $\mathbf{X} = [\mathbf{X}_1, \mathbf{X}_2]$ and $\boldsymbol{\beta} = [\boldsymbol{\beta}_1^\top, \boldsymbol{\beta}_2^\top]^\top$.

$$F = \frac{(\text{SSE(Model with } \mathbf{X}_1) - \text{SSE(Full)})/(p-k)}{\text{SSE(Full)}/(n-p-1)}.$$

Test for Individual Coefficients (t-Test)

Is the j-th predictor significant?

$$H_0 : \beta_j = c \quad \text{(typically } c = 0).$$

This is a special case of the General Linear Hypothesis with $\mathbf{A} = \boldsymbol{e}_j^\top$ (selecting the j-th element) and $\mathbf{b} = c$. From the F-test framework ($m = 1$):

$$F = \frac{(\hat{\beta}_j - c)\left[(\mathbf{X}^\top\mathbf{X})^{-1}_{jj}\right]^{-1}(\hat{\beta}_j - c)}{1 \cdot \hat{\sigma}^2} = \frac{(\hat{\beta}_j - c)^2}{\hat{\sigma}^2(\mathbf{X}^\top\mathbf{X})^{-1}_{jj}}.$$

Taking the square root (preserving sign):

$$t = \frac{\hat{\beta}_j - c}{\text{se}(\hat{\beta}_j)} \sim t_{n-p-1}, \quad \text{where se}(\hat{\beta}_j) = \hat{\sigma}\sqrt{(\mathbf{X}^\top\mathbf{X})^{-1}_{jj}}.$$

Remark 1.18 $F_{1,\nu} = t_\nu^2$. The F-test tests $\beta_j \neq c$ (two-sided), while the t-test can also test one-sided hypotheses ($\beta_j > c$).

Test for Linear Combinations

Does a linear combination equal a constant (e.g., $\beta_1 + \beta_2 = 1$)?

$$H_0 : \mathbf{c}^\top\boldsymbol{\beta} = d.$$

Using the same logic as the t-test:

$$t = \frac{\mathbf{c}^\top\hat{\boldsymbol{\beta}} - d}{\hat{\sigma}\sqrt{\mathbf{c}^\top(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{c}}} \sim t_{n-p-1}.$$

Test Name	Hypothesis (H_0)	Statistic	Distribution
General Linear	$\mathbf{A}\boldsymbol{\beta} = \mathbf{b}$	$F = \frac{(\text{SSE}_R - \text{SSE}_F)/m}{\hat{\sigma}^2}$	$F_{m,n-p-1}$
Global Significance	$\beta_1 = \cdots = \beta_p = 0$	$F = \frac{\text{SSR}/p}{\text{SSE}/(n-p-1)}$	$F_{p,n-p-1}$
Single Coefficient	$\beta_j = c$	$t = \frac{\hat{\beta}_j - c}{\text{se}(\hat{\beta}_j)}$	t_{n-p-1}
Linear Combination	$\mathbf{c}^\top \boldsymbol{\beta} = d$	$t = \frac{\mathbf{c}^\top \hat{\boldsymbol{\beta}} - d}{\text{se}(\mathbf{c}^\top \hat{\boldsymbol{\beta}})}$	t_{n-p-1}

Table 1.1: Summary of Hypothesis Tests in Linear Models

1.2.4 Summary of Tests

§ 1.3

Multicollinearity and Ridge Regression

1.3.1 Standardized Regression Model

Definition 1.19 (Standardization) Let Y_i and X_{ij} be the original observations. We define the centering and scaling factors as:

- **Centering:** $\bar{Y} = \frac{1}{n}\sum Y_i$ and $\bar{X}_j = \frac{1}{n}\sum_i X_{ij}$.
- **Scaling (Norm):** $L_{yy} = \sum_{i=1}^n (Y_i - \bar{Y})^2$ and $L_{jj} = \sum_{i=1}^n (X_{ij} - \bar{X}_j)^2$.

The standardized variables are defined as:

$$y_i = \frac{Y_i - \bar{Y}}{\sqrt{L_{yy}}}, \quad x_{ij} = \frac{X_{ij} - \bar{X}_j}{\sqrt{L_{jj}}}.$$

Matrix Formulation

We define the standardized vectors and matrices:

$$\boldsymbol{y} = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}, \quad \mathbf{X} = \begin{pmatrix} x_{11} & \cdots & x_{1p} \\ \vdots & \ddots & \vdots \\ x_{n1} & \cdots & x_{np} \end{pmatrix}, \quad \boldsymbol{\beta}^* = \begin{pmatrix} \beta_1^* \\ \vdots \\ \beta_p^* \end{pmatrix}.$$

The standardized linear regression model is:

$$\boldsymbol{y} = \mathbf{X}\boldsymbol{\beta}^* + \boldsymbol{\varepsilon}^*.$$

Note that no intercept term β_0 is included because both $\boldsymbol{y}$ and the columns of $\mathbf{X}$ are centered (mean zero).

Key Properties

The standardized design matrix $\mathbf{X}$ possesses elegant algebraic properties that simplify the OLS and Ridge solutions.

Proposition 1.20 Properties of $\mathbf{X}$:

1. **Zero Mean (Centering):** The columns sum to zero.

$$\mathbf{1}^\top \boldsymbol{y} = 0, \quad \mathbf{1}^\top \mathbf{X} = \mathbf{0}^\top.$$

   This implies $\mathbf{X}$ is orthogonal to the intercept vector.

2. **Unit Norm:** The diagonal elements of the cross-product matrix are all 1.

$$(\mathbf{X}^\top \mathbf{X})_{jj} = \sum_{i=1}^{n} x_{ij}^2 = 1.$$

3. **Correlation Structure:** The Gram matrix is exactly the sample correlation matrix of the predictors.

$$\mathbf{X}^\top \mathbf{X} = \mathbf{R}_{xx} = \begin{pmatrix} 1 & r_{12} & \cdots & r_{1p} \\ r_{21} & 1 & \cdots & r_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ r_{p1} & r_{p2} & \cdots & 1 \end{pmatrix}.$$

4. **Predictor-Response Correlation:** The vector $\mathbf{X}^\top \boldsymbol{y}$ contains the sample correlations between predictors and the response.

$$\mathbf{X}^\top \boldsymbol{y} = \boldsymbol{r}_{xy} = \left(r_{y1}, r_{y2}, \ldots, r_{yp} \right)^\top.$$

Relationship to Original Estimators

The standardized OLS estimator is given by:

$$\hat{\boldsymbol{\beta}}^* = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{y} = \mathbf{R}_{xx}^{-1} \boldsymbol{r}_{xy}.$$

Once $\hat{\boldsymbol{\beta}}^*$ is obtained, the regression coefficients $\hat{\beta}_j$ for the original unscaled data can be recovered via:

$$\hat{\beta}_j = \hat{\beta}_j^* \cdot \frac{\sqrt{L_{yy}}}{\sqrt{L_{jj}}}, \quad j = 1, \ldots, p.$$

The intercept $\hat{\beta}_0$ is recovered as:

$$\hat{\beta}_0 = \bar{Y} - \sum_{j=1}^{p} \hat{\beta}_j \bar{X}_j.$$

1.3.2 Multicollinearity

- **Perfect Multicollinearity:** The columns of $\mathbf{X}$ are linearly dependent, i.e., $\operatorname{rank}(\mathbf{X}) < p$. $\mathbf{X}^\top\mathbf{X}$ is singular; $\hat{\boldsymbol{\beta}}_{OLS}$ is undefined.
- **Near Multicollinearity:** There exists a vector $\boldsymbol{c} \neq \mathbf{0}$ such that $\mathbf{X}\boldsymbol{c} \approx \mathbf{0}$. $\mathbf{X}^\top\mathbf{X}$ is invertible but ill-conditioned.

Theorem 1.21 (Variance Inflation) Let $\lambda_1 \geqslant \lambda_2 \geqslant \cdots \geqslant \lambda_p > 0$ be the eigenvalues of $\mathbf{X}^\top\mathbf{X}$. The total variance of the OLS estimator is:

$$\sum_{j=1}^{p} \operatorname{Var}(\hat{\beta}_j) = \sigma^2 \operatorname{tr}((\mathbf{X}^\top\mathbf{X})^{-1}) = \sigma^2 \sum_{j=1}^{p} \frac{1}{\lambda_j}.$$

Proof. Using the spectral decomposition, $\mathbf{X}^\top\mathbf{X} = \mathbf{P}\boldsymbol{\Lambda}\mathbf{P}^\top$, where $\mathbf{P}$ is orthogonal and $\boldsymbol{\Lambda} = \operatorname{diag}(\lambda_1, \ldots, \lambda_p)$. The covariance matrix is $\operatorname{Var}(\hat{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1} = \sigma^2\mathbf{P}\boldsymbol{\Lambda}^{-1}\mathbf{P}^\top$. The total variance is the trace of this matrix:

$$\operatorname{tr}(\operatorname{Var}(\hat{\boldsymbol{\beta}})) = \sigma^2 \operatorname{tr}(\mathbf{P}\boldsymbol{\Lambda}^{-1}\mathbf{P}^\top) = \sigma^2 \operatorname{tr}(\boldsymbol{\Lambda}^{-1}\mathbf{P}^\top\mathbf{P}) = \sigma^2 \operatorname{tr}(\boldsymbol{\Lambda}^{-1}) = \sigma^2 \sum_{j=1}^{p} \frac{1}{\lambda_j}.$$

□

Corollary 1.22 If the condition number $\kappa = \lambda_1/\lambda_p$ is large (indicating multicollinearity), then small λ_p implies λ_p^{-1} is huge, causing the variance of $\hat{\boldsymbol{\beta}}$ to explode. This makes OLS estimates unstable and unreliable.

1.3.3 Ridge Regression

To mitigate variance inflation, Ridge Regression introduces a small bias to significantly reduce variance.

Definition 1.23 (Ridge Estimator) The Ridge estimator $\hat{\boldsymbol{\beta}}_R(k)$ is defined for a penalty parameter $k \geqslant 0$:

$$\hat{\boldsymbol{\beta}}_R(k) = (\mathbf{X}^\top\mathbf{X} + k\mathbf{I})^{-1}\mathbf{X}^\top\boldsymbol{y}.$$

Theorem 1.24 (Geometric Interpretation) The ridge estimator solves the constrained optimization problem:

$$\min_{\boldsymbol{\beta}} \|\tilde{\mathbf{Y}} - \tilde{\mathbf{X}}\boldsymbol{\beta}\|^2 \quad \text{subject to} \quad \|\boldsymbol{\beta}\|^2 \leqslant t$$

where k is the Lagrange multiplier corresponding to the constraint $\|\boldsymbol{\beta}\|^2 \leqslant t$.

Proof. The Lagrangian is:

$$\mathcal{L}(\boldsymbol{\beta}, \lambda) = \|\tilde{\mathbf{Y}} - \tilde{\mathbf{X}}\boldsymbol{\beta}\|^2 + \lambda(\|\boldsymbol{\beta}\|^2 - t)$$

Taking derivatives:

$$\frac{\partial \mathcal{L}}{\partial \boldsymbol{\beta}} = -2\tilde{\mathbf{X}}^T(\tilde{\mathbf{Y}} - \tilde{\mathbf{X}}\boldsymbol{\beta}) + 2\lambda\boldsymbol{\beta} = \mathbf{0}$$

Solving gives $\hat{\boldsymbol{\beta}}(\lambda) = (\tilde{\mathbf{X}}^T\tilde{\mathbf{X}} + \lambda\mathbf{I})^{-1}\tilde{\mathbf{X}}^T\tilde{\mathbf{Y}}$. □

1.3.4 Properties of the Ridge Estimator

We analyze the statistical properties of the Ridge estimator $\hat{\boldsymbol{\beta}}_R(k)$ relative to the true parameter $\boldsymbol{\beta}$. Recall the definition:

$$\hat{\boldsymbol{\beta}}_R(k) = \mathbf{W}_k\mathbf{X}^\top \boldsymbol{y}, \quad \text{where } \mathbf{W}_k = (\mathbf{X}^\top\mathbf{X} + k\mathbf{I})^{-1}.$$

Bias-Variance Decomposition

Unlike the OLS estimator, the Ridge estimator is biased. We derive its bias and variance explicitly.

Theorem 1.25 Bias and Variance:

1. **Bias:** The estimator shrinks the true parameter vector towards zero:

$$\text{Bias}(\hat{\boldsymbol{\beta}}_R(k)) = -k(\mathbf{X}^\top\mathbf{X} + k\mathbf{I})^{-1}\boldsymbol{\beta}.$$

2. **Variance:** The covariance matrix is given by:

$$\text{Var}(\hat{\boldsymbol{\beta}}_R(k)) = \sigma^2\mathbf{W}_k\mathbf{X}^\top\mathbf{X}\mathbf{W}_k.$$

Proof. 1. Bias Derivation: Substitute the model $\boldsymbol{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ into the estimator:

$$\begin{aligned}
\mathbb{E}[\hat{\boldsymbol{\beta}}_R(k)] &= \mathbb{E}[\mathbf{W}_k\mathbf{X}^\top(\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon})] \\
&= \mathbf{W}_k\mathbf{X}^\top\mathbf{X}\boldsymbol{\beta} + \mathbf{W}_k\mathbf{X}^\top\mathbb{E}[\boldsymbol{\varepsilon}] \quad (\text{since } \mathbb{E}[\boldsymbol{\varepsilon}] = \mathbf{0}) \\
&= (\mathbf{X}^\top\mathbf{X} + k\mathbf{I})^{-1}(\mathbf{X}^\top\mathbf{X})\boldsymbol{\beta}.
\end{aligned}$$

To simplify, add and subtract $k\mathbf{I}$ inside the second term:

$$\begin{aligned}
\mathbb{E}[\hat{\boldsymbol{\beta}}_R(k)] &= (\mathbf{X}^\top\mathbf{X} + k\mathbf{I})^{-1}(\mathbf{X}^\top\mathbf{X} + k\mathbf{I} - k\mathbf{I})\boldsymbol{\beta} \\
&= \left[\mathbf{I} - k(\mathbf{X}^\top\mathbf{X} + k\mathbf{I})^{-1}\right]\boldsymbol{\beta} \\
&= \boldsymbol{\beta} - k(\mathbf{X}^\top\mathbf{X} + k\mathbf{I})^{-1}\boldsymbol{\beta}.
\end{aligned}$$

Thus, Bias $= \mathbb{E}[\hat{\boldsymbol{\beta}}_R(k)] - \boldsymbol{\beta} = -k(\mathbf{X}^\top\mathbf{X} + k\mathbf{I})^{-1}\boldsymbol{\beta}$.

2. Variance Derivation: Using the property $\text{Var}(\mathbf{A}\boldsymbol{y}) = \mathbf{A}\,\text{Var}(\boldsymbol{y})\mathbf{A}^\top$:

$$\begin{aligned}
\text{Var}(\hat{\boldsymbol{\beta}}_R(k)) &= \text{Var}(\mathbf{W}_k\mathbf{X}^\top\boldsymbol{y}) \\
&= (\mathbf{W}_k\mathbf{X}^\top)(\sigma^2\mathbf{I})(\mathbf{W}_k\mathbf{X}^\top)^\top \\
&= \sigma^2\mathbf{W}_k\mathbf{X}^\top\mathbf{X}\mathbf{W}_k^\top.
\end{aligned}$$

Since $\mathbf{X}^\top\mathbf{X}$ and $\mathbf{W}_k$ are symmetric, $\mathbf{W}_k^\top = \mathbf{W}_k$. The result follows. □

Mean Squared Error (MSE) Analysis

The Mean Squared Error (MSE) combines bias and variance to assess the total estimation quality. To evaluate it analytically, we use the **Canonical Form** (Spectral Decomposition).

Let $\mathbf{X}^\top\mathbf{X} = \mathbf{P}\mathbf{\Lambda}\mathbf{P}^\top$ be the eigendecomposition, where $\mathbf{P}$ is orthogonal ($\mathbf{P}^\top\mathbf{P} = \mathbf{I}$) and $\mathbf{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_p)$. We rotate the coordinate system to the principal components:

$$\boldsymbol{\alpha} = \mathbf{P}^\top\boldsymbol{\beta} \quad \text{(Canonical parameters)}.$$

Theorem 1.26 (Scalar MSE Formula) The MSE of the Ridge estimator decomposes into a sum of component-wise Mean Squared Errors:

$$\text{MSE}(k) = \sigma^2 \sum_{j=1}^{p} \frac{\lambda_j}{(\lambda_j + k)^2} + k^2 \sum_{j=1}^{p} \frac{\alpha_j^2}{(\lambda_j + k)^2}.$$

Proof. The total MSE is the trace of the variance covariance matrix plus the squared norm of the bias vector:

$$\text{MSE}(k) = \text{tr}(\text{Var}(\hat{\boldsymbol{\beta}}_R(k))) + ||\text{Bias}(\hat{\boldsymbol{\beta}}_R(k))||^2.$$

1. Variance Term (Sum of Variances): Using the spectral decomposition, $\mathbf{W}_k = \mathbf{P}(\mathbf{\Lambda} + k\mathbf{I})^{-1}\mathbf{P}^\top$.

$$\begin{aligned}
\text{tr}(\text{Var}) &= \sigma^2 \,\text{tr}\left(\mathbf{P}(\mathbf{\Lambda} + k\mathbf{I})^{-1}\mathbf{P}^\top \cdot \mathbf{P}\mathbf{\Lambda}\mathbf{P}^\top \cdot \mathbf{P}(\mathbf{\Lambda} + k\mathbf{I})^{-1}\mathbf{P}^\top\right) \\
&= \sigma^2 \,\text{tr}\left((\mathbf{\Lambda} + k\mathbf{I})^{-1}\mathbf{\Lambda}(\mathbf{\Lambda} + k\mathbf{I})^{-1}\right) \quad \text{(by cyclicity of trace)} \\
&= \sigma^2 \sum_{j=1}^{p} \frac{\lambda_j}{(\lambda_j + k)^2}.
\end{aligned}$$

2. Bias Term (Squared Bias):

$$\begin{aligned}
||\text{Bias}||^2 &= ||-k\mathbf{P}(\mathbf{\Lambda} + k\mathbf{I})^{-1}\mathbf{P}^\top\boldsymbol{\beta}||^2 \\
&= k^2\boldsymbol{\beta}^\top\mathbf{P}(\mathbf{\Lambda} + k\mathbf{I})^{-1}(\mathbf{\Lambda} + k\mathbf{I})^{-1}\mathbf{P}^\top\boldsymbol{\beta} \\
&= k^2\boldsymbol{\alpha}^\top(\mathbf{\Lambda} + k\mathbf{I})^{-2}\boldsymbol{\alpha} \quad (\text{substituting } \boldsymbol{\alpha} = \mathbf{P}^\top\boldsymbol{\beta}) \\
&= k^2 \sum_{j=1}^{p} \frac{\alpha_j^2}{(\lambda_j + k)^2}.
\end{aligned}$$

Adding the two terms yields the theorem. □

The Existence Theorem

The trade-off between variance reduction and bias introduction leads to the most important theoretical justification for Ridge Regression.

Theorem 1.27 (Existence of Superior Ridge Estimator) Provided that $\boldsymbol{\beta} \neq \mathbf{0}$ and $\sigma^2 > 0$, there exists a constant $k > 0$ such that:

$$\text{MSE}[\hat{\boldsymbol{\beta}}_R(k)] < \text{MSE}[\hat{\boldsymbol{\beta}}_{OLS}].$$

Proof. Let $f(k) = \mathrm{MSE}(k)$. Note that $f(0) = \mathrm{MSE}_{OLS}$. We examine the derivative of $f(k)$ at $k = 0$. Differentiating the MSE formula term by term:

$$f'(k) = \frac{d}{dk}\left(\sum_{j=1}^{p} \frac{\sigma^2\lambda_j + k^2\alpha_j^2}{(\lambda_j + k)^2}\right) = \sum_{j=1}^{p} h_j'(k).$$

Let's compute the derivative for the j-th term $h_j(k)$:

$$\begin{aligned}
h_j'(k) &= \frac{(\lambda_j + k)^2(2k\alpha_j^2) - (\sigma^2\lambda_j + k^2\alpha_j^2)\cdot 2(\lambda_j + k)}{(\lambda_j + k)^4} \\
&= \frac{2(\lambda_j + k)[k\alpha_j^2(\lambda_j + k) - (\sigma^2\lambda_j + k^2\alpha_j^2)]}{(\lambda_j + k)^4} \\
&= \frac{2[k\alpha_j^2\lambda_j + k^2\alpha_j^2 - \sigma^2\lambda_j - k^2\alpha_j^2]}{(\lambda_j + k)^3} \\
&= \frac{2(k\alpha_j^2\lambda_j - \sigma^2\lambda_j)}{(\lambda_j + k)^3}.
\end{aligned}$$

Now, evaluate this derivative at $k = 0$:

$$f'(0) = \sum_{j=1}^{p} h_j'(0) = \sum_{j=1}^{p} \frac{2(0 - \sigma^2\lambda_j)}{\lambda_j^3} = -2\sigma^2 \sum_{j=1}^{p} \frac{1}{\lambda_j^2}.$$

Since $\sigma^2 > 0$ and $\lambda_j > 0$, we have $f'(0) < 0$. This implies that the MSE function is strictly decreasing in a neighborhood to the right of 0. Therefore, there exists some $k > 0$ where the MSE is lower than at $k = 0$. □

Chapter 2

Generalized Linear Models (GLM)

§ 2.1

The Exponential Family and GLM Structure

Generalized Linear Models (GLMs) extend the traditional linear regression framework to allow for non-normal response variables and non-constant variance.

2.1.1 The Exponential Family

The foundation of GLMs is the exponential family of distributions.

Definition 2.1 (Exponential Family) A random variable Y belongs to the exponential family if its probability density (or mass) function can be written in the canonical form:

$$f(y \mid \theta, \phi) = \exp\left\{\frac{y\theta - b(\theta)}{a(\phi)} + c(y, \phi)\right\} \tag{2.1}$$

where:

- θ: **Natural parameter** (canonical parameter). It relates directly to the mean μ.
- ϕ: **Dispersion parameter**. It relates to the variance (e.g., σ^2 in Normal distribution).
- $b(\theta)$: **Cumulant function**. Its derivatives generate the moments of the distribution.
- $a(\phi)$: Dispersion function, typically $a(\phi) = \phi/\omega$ (where ω is a known weight, often 1).

Theorem 2.2 (Bartlett's Identities) Under regularity conditions (permitting the interchange of differentiation and integration), the moments of Y satisfy:

$$\mathbb{E}[Y] = \mu = b'(\theta) \tag{2.2}$$

$$\mathrm{Var}[Y] = b''(\theta)a(\phi) \tag{2.3}$$

Here, $V(\mu) = b''(\theta(\mu))$ is called the **Variance Function**, which characterizes how the variance depends on the mean.

Proof. 1. Mean: Since f is a density, $\int f(y \mid \theta, \phi)\mathrm{d}y = 1$. Differentiating both sides with respect to θ:

$$\int \frac{\partial}{\partial\theta} \exp\left\{\frac{y\theta - b(\theta)}{a(\phi)} + c\right\} \mathrm{d}y = \int \frac{y - b'(\theta)}{a(\phi)} f(y)\mathrm{d}y = 0.$$

Multiplying by $a(\phi)$, we obtain $\mathbb{E}[Y - b'(\theta)] = 0$, which implies $\mathbb{E}[Y] = b'(\theta)$.

2. Variance: Differentiate the score expectation $\int (y - b'(\theta)) f(y)\mathrm{d}y = 0$ with respect to θ again:

$$\int \left[-b''(\theta)f(y) + (y - b'(\theta))\frac{y - b'(\theta)}{a(\phi)} f(y)\right] \mathrm{d}y = 0.$$

Rearranging terms yields:

$$-b''(\theta) \underbrace{\int f(y)\mathrm{d}y}_{1} + \frac{1}{a(\phi)} \underbrace{\int (y - \mathbb{E}[Y])^2 f(y)\mathrm{d}y}_{\mathrm{Var}[Y]} = 0.$$

Thus, $\mathrm{Var}[Y] = b''(\theta)a(\phi)$. □

2.1.2 Generalized Linear Models (GLM)

Definition 2.3 (GLM Components) A Generalized Linear Model consists of three independent components:

1. **Random Component:** The response variables $Y_1, \ldots, Y_n$ are independent and come from an exponential family distribution with mean $\mathbb{E}[Y_i] = \mu_i$.

2. **Systematic Component:** A linear predictor η_i is formed by the linear combination of predictors $\mathbf{x}_i$ and parameters $\boldsymbol{\beta}$:

$$\eta_i = \mathbf{x}_i^\top \boldsymbol{\beta}.$$

3. **Link Function:** A monotonic, differentiable function $g(\cdot)$ establishes the relationship between the random and systematic components:

$$g(\mu_i) = \eta_i \quad \Longleftrightarrow \quad \mu_i = g^{-1}(\eta_i).$$

Definition 2.4 (Canonical Link) A link function is called the **canonical link** if it links the linear predictor directly to the natural parameter:

$$\eta = \theta \quad \Longrightarrow \quad g(\mu) = \theta(\mu).$$

Use of the canonical link leads to simplified sufficient statistics ($\mathbf{X}^\top\mathbf{Y}$) and guarantees the convexity of the negative log-likelihood (ensuring a unique MLE).

2.1.3 Summary of Common Distributions

The following table summarizes the key components for common exponential family members. Note that for GLMs, the **Variance Function** $V(\mu)$ is crucial as it dictates the heteroscedasticity structure.

Table 2.1: Structure of Exponential Family Distributions in GLM

Distribution	**Natural Param**	**Cumulant**	**Mean**	**Variance Func**	**Canonical Link**
$f(y)$	θ	$b(\theta)$	$\mu = b'(\theta)$	$V(\mu) = b''(\theta)$	$g(\mu) = \theta$
Normal (σ^2 known)	μ	$\frac{\theta^2}{2}$	θ	1	μ (Identity)
Poisson	$\ln \lambda$	e^θ	e^θ	μ	$\ln \mu$ (Log)
Bernoulli	$\ln \frac{p}{1-p}$	$\ln(1+e^\theta)$	$\frac{e^\theta}{1+e^\theta}$	$\mu(1-\mu)$	$\ln \frac{\mu}{1-\mu}$ (Logit)
Gamma	$-\frac{1}{\mu}$	$-\ln(-\theta)$	$-\frac{1}{\theta}$	μ^2	$-\frac{1}{\mu}$ (Reciprocal)
Inverse Gaussian	$-\frac{1}{2\mu^2}$	$-\sqrt{-2\theta}$	$(-2\theta)^{-1/2}$	μ^3	$-\frac{1}{2\mu^2}$ (Squared Recip.)

§ 2.2

Parameter Estimation Methods

The Maximum Likelihood Estimator (MLE) for GLMs is obtained by maximizing the log-likelihood function. Unlike the OLS case for Normal error models, the likelihood equations for GLMs are generally non-linear in the parameters $\boldsymbol{\beta}$. Therefore, we must rely on iterative numerical algorithms.

2.2.1 Likelihood and Score Equations

Consider n observations Y_i from an exponential family with density $f(y_i|\theta_i, \phi)$. The log-likelihood is:

$$\ell(\boldsymbol{\beta}) = \sum_{i=1}^{n} \ell_i(\boldsymbol{\beta}) = \sum_{i=1}^{n} \left[\frac{y_i\theta_i - b(\theta_i)}{a(\phi)} + c(y_i, \phi) \right].$$

The parameters $\boldsymbol{\beta}$ enter the equation through the linear predictor $\eta_i = \mathbf{x}_i^\top \boldsymbol{\beta}$ and the link function $\eta_i = g(\mu_i)$.

Theorem 2.5 (The Score Vector) The gradient of the log-likelihood, known as the **Score Vector $\boldsymbol{U}(\boldsymbol{\beta}) = \nabla_{\boldsymbol{\beta}} \ell(\boldsymbol{\beta})$**, is given by:

$$\boldsymbol{U}(\boldsymbol{\beta}) = \mathbf{X}^\top \mathbf{W}\mathbf{G}(\boldsymbol{y} - \boldsymbol{\mu}),$$

where:

- $\mathbf{W} = \mathrm{diag}(w_i)$ is the weight matrix with $w_i = \frac{1}{a(\phi)V(\mu_i)[g'(\mu_i)]^2}$.
- $\mathbf{G} = \mathrm{diag}(g'(\mu_i))$ contains the derivatives of the link function.

In component form, the j-th element is:

$$U_j(\boldsymbol{\beta}) = \sum_{i=1}^{n} \frac{(y_i - \mu_i)}{a(\phi)V(\mu_i)} \frac{x_{ij}}{g'(\mu_i)}.$$

Proof. By the Chain Rule, for each parameter β_j:

$$\begin{aligned} \frac{\partial \ell_i}{\partial \beta_j} &= \frac{\partial \ell_i}{\partial \theta_i} \cdot \frac{\partial \theta_i}{\partial \mu_i} \cdot \frac{\partial \mu_i}{\partial \eta_i} \cdot \frac{\partial \eta_i}{\partial \beta_j} \\ &= \frac{y_i - \mu_i}{a(\phi)} \cdot \frac{1}{V(\mu_i)} \cdot \frac{1}{g'(\mu_i)} \cdot x_{ij} \end{aligned}$$

□

Unlike Ordinary Least Squares (OLS), where the normal equations linear in $\boldsymbol{\beta}$ yield a closed-form solution $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\boldsymbol{y}$, the score equations $\boldsymbol{U}(\boldsymbol{\beta}) = \mathbf{0}$ in GLMs are generally **non-linear** in $\boldsymbol{\beta}$. This is because the mean μ_i is connected to the parameters via the non-linear link function $g(\cdot)$.

Consequently, no explicit analytical solution exists. We must rely on **iterative numerical algorithms** (like Newton's method) to approximate the roots. These algorithms typically linearize the score function via Taylor expansion, which requires computing the gradient of the score vector—i.e., the second derivatives of the log-likelihood. This necessitates the derivation of the **Hessian** and **Fisher Information** matrices.

2.2.2 Hessian and Fisher Information

To employ Newton-type optimization algorithms, we require the second derivatives of the log-likelihood.

- The **Observed Information Matrix** (Hessian) is $\mathbf{H}(\boldsymbol{\beta}) = \nabla^2 \ell(\boldsymbol{\beta})$.
- The **Fisher Information Matrix** is $\mathcal{I}(\boldsymbol{\beta}) = \mathbb{E}[-\mathbf{H}(\boldsymbol{\beta})] = \mathbb{E}[\boldsymbol{U}(\boldsymbol{\beta})\boldsymbol{U}(\boldsymbol{\beta})^\top]$.

To understand the difference between Newton-Raphson and Fisher Scoring, we first derive the exact Hessian. Recall the score component for the j-th parameter:

$$U_j(\boldsymbol{\beta}) = \sum_{i=1}^{n} \frac{y_i - \mu_i}{a(\phi)V(\mu_i)g'(\mu_i)} x_{ij}.$$

Let $w_i^* = \frac{1}{a(\phi)V(\mu_i)g'(\mu_i)}$. Using the product rule to differentiate U_j with respect to β_k:

$$\frac{\partial U_j}{\partial \beta_k} = \sum_{i=1}^{n} \left[\frac{\partial (y_i - \mu_i)}{\partial \beta_k} w_i^* + (y_i - \mu_i) \frac{\partial w_i^*}{\partial \beta_k} \right] x_{ij}.$$

Using the chain rule $\frac{\partial \mu_i}{\partial \beta_k} = \frac{1}{g'(\mu_i)} x_{ik}$, the (j,k)-th element of the Hessian is:

$$H_{jk} = \underbrace{-\sum_{i=1}^{n} \frac{1}{a(\phi)V(\mu_i)[g'(\mu_i)]^2} x_{ij} x_{ik}}_{\text{Term 1: Deterministic Part}} + \underbrace{\sum_{i=1}^{n} (y_i - \mu_i) \frac{\partial w_i^*}{\partial \beta_k} x_{ij}}_{\text{Term 2: Random Part}}.$$

This derivation highlights the key difference between optimization methods:

- **Newton-Raphson** uses the exact Hessian $\mathbf{H}$. Calculation of Term 2 is computationally expensive (involving derivatives of $V(\cdot)$ and $g'(\cdot)$) and depends on the residuals $(y_i - \mu_i)$. If the model fits poorly, Term 2 can dominate, potentially making $\mathbf{H}$ non-negative definite.
- **Fisher Scoring** uses $\mathcal{I}(\boldsymbol{\beta}) = \mathbb{E}[-\mathbf{H}(\boldsymbol{\beta})]$. Since $\mathbb{E}[y_i - \mu_i] = 0$, the expectation of Term 2 vanishes exactly.

Thus, Fisher Scoring discards the "noise" from the observed residuals, relying instead on the expected curvature:

$$(\mathcal{I})_{jk} = -\mathbb{E}[\text{Term 1}] = \sum_{i=1}^{n} \frac{1}{a(\phi)V(\mu_i)[g'(\mu_i)]^2} x_{ij} x_{ik}.$$

Theorem 2.6 The Fisher Information matrix for a GLM is given by:

$$\mathcal{I}(\boldsymbol{\beta}) = \mathbf{X}^\top \mathbf{W} \mathbf{X},$$

where $\mathbf{W} = \text{diag}(w_i)$.

Proof. The proof follows directly from the Hessian derivation above. □

Remark 2.7 When the canonical link is used (i.e., $\eta = \theta$):

- The weights simplify significantly to $w_i = V(\mu_i)/a(\phi)$.
- In this specific case, the Observed Hessian equals Expected Fisher Information, meaning Newton-Raphson and Fisher Scoring are identical algorithms.

2.2.3 Optimization Algorithms

We wish to solve $\boldsymbol{U}(\boldsymbol{\beta}) = \mathbf{0}$.

1. **Newton-Raphson Method:**

$$\boldsymbol{\beta}^{(t+1)} = \boldsymbol{\beta}^{(t)} - [\mathbf{H}(\boldsymbol{\beta}^{(t)})]^{-1} \boldsymbol{U}(\boldsymbol{\beta}^{(t)}).$$

   Drawback: **H** is computationally expensive to calculate and may not be negative definite (can lead to divergence).

2. **Fisher Scoring Method:** Replaces the observed Hessian **H** with the expected Fisher Information $-\mathcal{I}$.

$$\boldsymbol{\beta}^{(t+1)} = \boldsymbol{\beta}^{(t)} + [\mathcal{I}(\boldsymbol{\beta}^{(t)})]^{-1} \boldsymbol{U}(\boldsymbol{\beta}^{(t)}).$$

   Advantage: $\mathcal{I}$ is positive semi-definite, ensuring the algorithm moves in an ascent direction.

Iteratively Reweighted Least Squares (IRLS)

We now demonstrate that the Fisher Scoring update is algebraically equivalent to solving a specific Weighted Least Squares (WLS) problem at each step.

Definition 2.8 (Working Variables) At iteration t, we define the **Working Response** (or Adjusted Dependent Variable) $z_i^{(t)}$ as:

$$z_i^{(t)} = \eta_i^{(t)} + (y_i - \mu_i^{(t)}) g'(\mu_i^{(t)}).$$

Geometrically, z_i represents the data y_i transformed onto the linear predictor scale via a first-order Taylor expansion.

Theorem 2.9 The Fisher Scoring update step is equivalent to solving the normal equations for a weighted regression of $\boldsymbol{z}^{(t)}$ on $\mathbf{X}$ with weights $\mathbf{W}^{(t)}$:

$$(\mathbf{X}^\top \mathbf{W}^{(t)} \mathbf{X}) \boldsymbol{\beta}^{(t+1)} = \mathbf{X}^\top \mathbf{W}^{(t)} \boldsymbol{z}^{(t)}.$$

Proof. Starting with the Fisher Scoring update equation:

$$\boldsymbol{\beta}^{(t+1)} = \boldsymbol{\beta}^{(t)} + \underbrace{(\mathbf{X}^\top \mathbf{W}^{(t)} \mathbf{X})}_{\mathcal{I}^{(t)}}{}^{-1} \boldsymbol{U}^{(t)}.$$

Premultiply both sides by the information matrix $\mathbf{X}^\top \mathbf{W}^{(t)} \mathbf{X}$:

$$(\mathbf{X}^\top \mathbf{W}^{(t)} \mathbf{X}) \boldsymbol{\beta}^{(t+1)} = (\mathbf{X}^\top \mathbf{W}^{(t)} \mathbf{X}) \boldsymbol{\beta}^{(t)} + \boldsymbol{U}^{(t)}.$$

Recall the matrix form of the score vector derived earlier: $\boldsymbol{U}^{(t)} = \mathbf{X}^\top \mathbf{W}^{(t)} \boldsymbol{r}^*$, where vector $\boldsymbol{r}^*$ has elements $(y_i - \mu_i^{(t)}) g'(\mu_i^{(t)})$.

Substituting this back into the update equation:

$$\begin{aligned}(\mathbf{X}^\top \mathbf{W}^{(t)} \mathbf{X}) \boldsymbol{\beta}^{(t+1)} &= \mathbf{X}^\top \mathbf{W}^{(t)} \underbrace{\mathbf{X} \boldsymbol{\beta}^{(t)}}_{\boldsymbol{\eta}^{(t)}} + \mathbf{X}^\top \mathbf{W}^{(t)} \boldsymbol{r}^* \\ &= \mathbf{X}^\top \mathbf{W}^{(t)} \left(\boldsymbol{\eta}^{(t)} + \boldsymbol{r}^* \right).\end{aligned}$$

The term in the parentheses is precisely the working response vector $\boldsymbol{z}^{(t)}$:

$$\boldsymbol{\eta}^{(t)} + \boldsymbol{r}^* = \boldsymbol{z}^{(t)}.$$

Thus, the update reduces to solving $(\mathbf{X}^\top \mathbf{W}^{(t)} \mathbf{X}) \boldsymbol{\beta}^{(t+1)} = \mathbf{X}^\top \mathbf{W}^{(t)} \boldsymbol{z}^{(t)}$. □

Algorithm 1 IRLS Algorithm for GLM Estimation

Input: Data $(\mathbf{X}, \boldsymbol{y})$, Distribution Family, Link Function $g(\cdot)$

1: **Initialize**:
2: $\boldsymbol{\mu}^{(0)} = \boldsymbol{y} + \delta$ (small constant), $\boldsymbol{\eta}^{(0)} = g(\boldsymbol{\mu}^{(0)})$
3: **Iterate** $t = 0, 1, 2, \ldots$ until convergence:
4: **Step 1: Compute Weights**

$$w_i^{(t)} = \frac{1}{a(\phi) V(\mu_i^{(t)}) [g'(\mu_i^{(t)})]^2}$$

5: **Step 2: Compute Working Response**

$$z_i^{(t)} = \eta_i^{(t)} + (y_i - \mu_i^{(t)}) g'(\mu_i^{(t)})$$

6: **Step 3: Solve Weighted Least Squares**

$$\boldsymbol{\beta}^{(t+1)} = (\mathbf{X}^\top \mathbf{W}^{(t)} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{W}^{(t)} \boldsymbol{z}^{(t)}$$

7: **Step 4: Update Predictors and Means**

$$\boldsymbol{\eta}^{(t+1)} = \mathbf{X} \boldsymbol{\beta}^{(t+1)}, \quad \boldsymbol{\mu}^{(t+1)} = g^{-1}(\boldsymbol{\eta}^{(t+1)})$$

8: **Check Convergence**: If $\|\boldsymbol{\beta}^{(t+1)} - \boldsymbol{\beta}^{(t)}\| < \epsilon$, stop.

2.2.4 Statistical Inference and Asymptotics

Once the MLE $\hat{\beta}$ is obtained, we rely on the large-sample properties of likelihood theory to construct confidence intervals and hypothesis tests.

Theorem 2.10 (Asymptotic Normality of $\hat{\beta}$) Under standard regularity conditions, as $n \to \infty$, the MLE $\hat{\beta}$ is consistent and asymptotically normal:

$$\sqrt{n}(\hat{\beta} - \beta) \xrightarrow{d} \mathcal{N}\left(\mathbf{0}, \mathcal{I}_1(\beta)^{-1}\right)$$

In finite samples, this implies $\hat{\beta} \approx \mathcal{N}\left(\beta, (\mathbf{X}^\top \mathbf{W}\mathbf{X})^{-1}\right)$.

Proof. Consider the first-order Taylor expansion of the score function $\boldsymbol{U}(\hat{\beta})$ around the true parameter β. Since $\hat{\beta}$ is the MLE, $\boldsymbol{U}(\hat{\beta}) = \mathbf{0}$, satisfying:

$$\mathbf{0} = \boldsymbol{U}(\beta) + \nabla^2 \ell(\beta^*)(\hat{\beta} - \beta)$$

where β^* lies between $\hat{\beta}$ and β. Rearranging and introducing the scaling factor $\sqrt{n}$ yields:

$$\sqrt{n}(\hat{\beta} - \beta) = \underbrace{\left[-\frac{1}{n}\nabla^2 \ell(\beta^*)\right]^{-1}}_{\text{(A)}} \underbrace{\left(\frac{1}{\sqrt{n}}\boldsymbol{U}(\beta)\right)}_{\text{(B)}}$$

As $n \to \infty$, we apply the following limit theorems:

(A) **Convergence of Hessian:** By the Weak Law of Large Numbers (WLLN) and the consistency of $\hat{\beta}$, the observed information converges to the expected Fisher information:

$$-\frac{1}{n}\nabla^2 \ell(\beta^*) \xrightarrow{p} \mathbb{E}\left[-\nabla^2 \ell(\beta)\right] = \mathcal{I}_1(\beta)$$

(B) **Distribution of Score:** By the Central Limit Theorem (CLT), since $\boldsymbol{U}(\beta)$ is a sum of independent zero-mean variables:

$$\frac{1}{\sqrt{n}}\boldsymbol{U}(\beta) \xrightarrow{d} \mathcal{N}\left(\mathbf{0}, \mathcal{I}_1(\beta)\right)$$

Applying **Slutsky's Theorem** to combine (A) and (B):

$$\sqrt{n}(\hat{\beta} - \beta) \xrightarrow{d} \mathcal{I}_1(\beta)^{-1} \cdot \mathcal{N}\left(\mathbf{0}, \mathcal{I}_1(\beta)\right) = \mathcal{N}\left(\mathbf{0}, \mathcal{I}_1(\beta)^{-1}\right)$$

For finite samples, substituting the total information $\mathcal{I}_n(\beta) = n\mathcal{I}_1(\beta) = \mathbf{X}^\top \mathbf{W}\mathbf{X}$, we obtain the approximation:

$$\hat{\beta} \sim \mathcal{N}\left(\beta, (\mathbf{X}^\top \mathbf{W}\mathbf{X})^{-1}\right)$$

□

Estimated Covariance Matrix

In practice, we estimate the covariance by substituting $\hat{\beta}$ and $\hat{\phi}$ into the information matrix.

$$\widehat{Var}(\hat{\beta}) = \hat{\phi}(X^\top \hat{W} X)^{-1}$$

Estimation of Dispersion Parameter ϕ

While ϕ cancels out in the calculation of $\hat{\beta}$ (for canonical exponential families), it acts as a scaling factor for the variance.

- **Method of Moments Estimator:** For families like Normal or Gamma, we use the generalized Pearson statistic. This is based on the moment condition $\mathbb{E}[(Y_i - \mu_i)^2] \approx \phi V(\mu_i)$.

  $$\hat{\phi} = \frac{1}{n-p} \sum_{i=1}^{n} \frac{(y_i - \hat{\mu}_i)^2}{V(\hat{\mu}_i)} = \frac{\chi^2_{Pearson}}{n-p}$$

  Using $n - p$ instead of n provides a bias correction for the estimation of β, similar to using $n - 1$ in sample variance.

- **Fixed Dispersion:** For Poisson ($\phi = 1$) and Bernoulli ($\phi = 1$), this step is usually skipped unless we are accounting for *Overdispersion* (where we might estimate a Quasi-Poisson $\hat{\phi} > 1$).

§ 2.3

General Hypothesis Testing Framework

Before deriving tests specific to GLMs, we review the three classical asymptotic tests for the general linear hypothesis. Consider partitioning the parameter vector $\boldsymbol{\beta}$ into nuisance parameters $\boldsymbol{\beta}_1$ (dimension $p - q$) and parameters of interest $\boldsymbol{\beta}_2$ (dimension q):

$$H_0 : \boldsymbol{\beta}_2 = \boldsymbol{\beta}_{20} \quad \text{vs} \quad H_1 : \boldsymbol{\beta}_2 \neq \boldsymbol{\beta}_{20}.$$

Let $\hat{\boldsymbol{\beta}}$ be the unrestricted MLE (under H_1) and $\tilde{\boldsymbol{\beta}}$ be the restricted MLE (under H_0).

2.3.1 The Three Classical Tests

The three tests approach the hypothesis from different geometric perspectives on the log-likelihood surface $\ell(\boldsymbol{\beta})$.

Likelihood Ratio Test (LRT)

Definition 2.11 The LRT measures the **vertical distance** between the log-likelihood peaks of the full and reduced models:

$$\lambda_{LR} = 2[\ell(\hat{\boldsymbol{\beta}}) - \ell(\tilde{\boldsymbol{\beta}})].$$

Theorem 2.12 Under H_0 and regularity conditions, $\lambda_{LR} \xrightarrow{d} \chi^2(q)$ as $n \to \infty$.

Sketch. Expand $\ell(\tilde{\boldsymbol{\beta}})$ around $\hat{\boldsymbol{\beta}}$ using a second-order Taylor series:

$$\ell(\tilde{\boldsymbol{\beta}}) \approx \ell(\hat{\boldsymbol{\beta}}) + (\tilde{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}})^\top \nabla \ell(\hat{\boldsymbol{\beta}}) - \frac{1}{2}(\tilde{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}})^\top \mathcal{I}(\hat{\boldsymbol{\beta}})(\tilde{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}).$$

Since $\hat{\boldsymbol{\beta}}$ is the MLE, $\nabla \ell(\hat{\boldsymbol{\beta}}) = \mathbf{0}$. Rearranging gives:

$$2[\ell(\hat{\boldsymbol{\beta}}) - \ell(\tilde{\boldsymbol{\beta}})] \approx (\tilde{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}})^\top \mathcal{I}(\hat{\boldsymbol{\beta}})(\tilde{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}).$$

This quadratic form of asymptotically normal vectors follows a $\chi^2(q)$ distribution. □

Wald Test

Definition 2.13 The Wald test measures the **horizontal distance** between the unrestricted estimate $\hat{\boldsymbol{\beta}}_2$ and the hypothesized value $\boldsymbol{\beta}_{20}$, standardized by the curvature (variance):

$$W = (\hat{\boldsymbol{\beta}}_2 - \boldsymbol{\beta}_{20})^\top \left[\widehat{\mathrm{Cov}}(\hat{\boldsymbol{\beta}}_2)\right]^{-1} (\hat{\boldsymbol{\beta}}_2 - \boldsymbol{\beta}_{20}).$$

Remark 2.14 (Computation) $\widehat{\mathrm{Cov}}(\hat{\boldsymbol{\beta}}_2)$ is the submatrix of the **inverse** Fisher Information $\mathcal{I}^{-1}$, specifically the block corresponding to $\boldsymbol{\beta}_2$.

Score Test (Rao)

Definition 2.15 The Score test measures the **slope (gradient)** of the log-likelihood at the null hypothesis $\tilde{\boldsymbol{\beta}}$. If H_0 is true, the gradient at the restricted optimum should be close to zero.

$$S = \boldsymbol{U}(\tilde{\boldsymbol{\beta}})^\top [\mathcal{I}(\tilde{\boldsymbol{\beta}})]^{-1} \boldsymbol{U}(\tilde{\boldsymbol{\beta}}).$$

Remark 2.16 (Computational Advantage) The Score test requires fitting **only the null model**. For GLMs with canonical links, it simplifies to:

$$S = (\boldsymbol{y} - \tilde{\boldsymbol{\mu}})^\top \mathbf{X}(\mathbf{X}^\top \tilde{\mathbf{W}} \mathbf{X})^{-1} \mathbf{X}^\top (\boldsymbol{y} - \tilde{\boldsymbol{\mu}}).$$

Theorem 2.17 (Asymptotic Equivalence) Under H_0, the three tests are asymptotically equivalent:

$$\lambda_{LR} = W + o_p(1) = S + o_p(1).$$

In finite samples, the order LRT $\geqslant$ Wald $\geqslant$ Score often holds, but depends on the shape of the likelihood.

Chapter 3

Regularization & Variable Selection

§ 3.1 Introduction: From Subset Selection to Convex Relaxation

3.1.1 Subset Selection Framework

Definition 3.1 (Subset Selection) For an index set $S \subseteq \{1, 2, \cdots, p\}$, define the corresponding submodel as:

$$Y = X_S \beta_S + \varepsilon$$

where X_S denotes the submatrix formed by columns indexed by S, and β_S is the corresponding coefficient vector.For subset S, we denote:

- $\mathrm{RSS}(S) = \|Y - X_S \hat{\beta}_S\|_2^2$
- $H_S = X_S (X_S^T X_S)^{-1} X_S^T$ (projection matrix)

Theorem 3.2 (Monotonicity of RSS) For nested subsets $S \subset T$, we have:

$$\mathrm{RSS}(S) \geqslant \mathrm{RSS}(T)$$

Equality holds if and only if the additional variables in $T \setminus S$ contribute no explanatory power.

Proof. Let H_S and H_T be the projection matrices for subsets S and T, respectively. Since $S \subset T$, we have $\mathrm{col}(X_S) \subseteq \mathrm{col}(X_T)$, and thus $H_T - H_S \geq 0$. Therefore:

$$\mathrm{RSS}(S) - \mathrm{RSS}(T) = Y^T (H_T - H_S) Y \geqslant 0$$

Equality holds when Y is orthogonal to $\mathrm{col}(X_T) \ominus \mathrm{col}(X_S)$. □

3.1.2 Computational Challenge & ℓ_0 Regularization

Definition 3.3 (ℓ_0 Regularization Equivalence) Optimal subset selection is equivalent to the following ℓ_0 regularization problem:

$$\min_{\beta \in \mathbb{R}^p} \frac{1}{2n} \|Y - X\beta\|_2^2 + \lambda \|\beta\|_0$$

where $\|\beta\|_0 = \sum_{j=1}^{p} \mathbb{I}(\beta_j \neq 0)$ is the ℓ_0 pseudo-norm.

Remark 3.4 (Motivation for Relaxation) While ℓ_0 regularization is theoretically appealing for variable selection, it is **computationally infeasible** for high-dimensional data (where $p \gg n$).

- The optimization problem is non-convex and **NP-hard**.
- Exhaustive search requires fitting 2^p models.

To address this, we consider **convex relaxations** (e.g., replacing ℓ_0 with ℓ_1) or other continuous penalty functions. The following sections explore these methods, starting with their behavior in a simplified geometric setting.

3.1.3 Classical Selection Algorithms

We introduce three representative classical algorithms in sequence, including an exhaustive optimal method and two greedy heuristic methods with computational efficiency.

Algorithm 2 Optimal Subset Selection

Input: Design matrix $X \in \mathbb{R}^{n\times p}$, response vector $Y \in \mathbb{R}^n$

1: Initialize null model M_0 with $S_0 = \varnothing$
2: **for** $k = 1$ to p **do**
3:     Compute all subsets of size k: $\mathcal{S}_k = \{S \subseteq \{1, \cdots, p\} : |S| = k\}$
4:     **for** each $S \in \mathcal{S}_k$ **do**
5:         Compute $\text{RSS}(S) = Y^T(I - H_S)Y$
6:     **end for**
7:     Select optimal subset at this size: $S_k = \arg\min_{S\in\mathcal{S}_k} \text{RSS}(S)$
8: **end for**
9: Select final model from $\{S_0, S_1, \cdots, S_p\}$:

$$S^* = \arg \min_{k\in\{0,1,\cdots,p\}} \text{Criterion}(S_k)$$

Output: Optimal subset S^*

3.1.4 Model Selection Criteria

To select the optimal model size (or regularization parameter λ), we typically use:

- **AIC**: $n \log(\text{RSS}/n) + 2|S|$ (Efficient, not consistent)
- **BIC**: $n \log(\text{RSS}/n) + |S| \log n$ (Consistent, not efficient)
- **Cross-Validation (CV)**: Estimates prediction error by splitting data into K folds.

Algorithm 3 Forward Selection

Input: $X \in \mathbb{R}^{n \times p}$, $Y \in \mathbb{R}^n$

Initialize: $S^{(0)} = \varnothing$, $\text{RSS}^{(0)} = Y^T Y$

for $m = 0$ to $p - 1$ **do**

For each $j \notin S^{(m)}$, compute:

$$\text{RSS}_j = \min_{\beta} \|Y - X_{S^{(m)} \cup \{j\}} \beta\|^2$$

Find $j^* = \arg\min_{j \notin S^{(m)}} \text{RSS}_j$

Update: $S^{(m+1)} = S^{(m)} \cup \{j^*\}$

Update: $\text{RSS}^{(m+1)} = \text{RSS}_{j^*}$

end for

Select final model from $\{S^{(0)}, S^{(1)}, \cdots, S^{(p)}\}$:

$$S^* = \arg \min_{k \in \{0,1,\cdots,p\}} \text{Criterion}(S_k)$$

Output: Optimal subset S^*

Algorithm 4 Backward Elimination

Input: $X \in \mathbb{R}^{n \times p}$, $Y \in \mathbb{R}^n$

Initialize: $S^{(0)} = \{1, 2, \cdots, p\}$

Compute full model: $\text{RSS}^{(0)} = \|Y - X\hat{\beta}\|^2$

for $m = 0$ to $p - 1$ **do**

For each $j \in S^{(m)}$, compute:

$$\text{RSS}_{-j} = \min_{\beta} \|Y - X_{S^{(m)} \setminus \{j\}} \beta\|^2$$

Find $j^* = \arg\min_{j \in S^{(m)}} \text{RSS}_{-j}$

Update: $S^{(m+1)} = S^{(m)} \setminus \{j^*\}$

Update: $\text{RSS}^{(m+1)} = \text{RSS}_{-j^*}$

end for

Select final model from $\{S^{(0)}, S^{(1)}, \cdots, S^{(p)}\}$:

$$S^* = \arg \min_{k \in \{0,1,\cdots,p\}} \text{Criterion}(S_k)$$

Output: Optimal subset S^*

§ 3.2

Intuitive Understanding under Orthogonal Design

Before discussing general solutions involving complex optimization (like Coordinate Descent), we first examine the behavior of different penalties under the simplified setting of **Orthogonal Design**. This provides clear geometric intuition into how each penalty "selects" or "shrinks" variables.

3.2.1 Problem Setup

Consider the penalized least squares problem:

$$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \left\{ \frac{1}{2}\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \sum_{j=1}^{p} P_\lambda(|\beta_j|) \right\} \tag{3.1}$$

Assume **Orthogonal Design**: $\frac{1}{n}\mathbf{X}^T\mathbf{X} = \mathbf{I}_p$.

Theorem 3.5 (Decomposition) Under orthogonal design, the problem decomposes into p independent univariate problems:

$$\hat{\beta}_j = \arg\min_{\beta_j} \left\{ \frac{1}{2}(z_j - \beta_j)^2 + P_\lambda(|\beta_j|) \right\}, \quad j = 1, \ldots, p \tag{3.2}$$

where $z_j = \mathbf{x}_j^T\mathbf{y}/n$ is the OLS estimate for the j-th variable.

Proof. The objective function can be rewritten as:

$$\begin{aligned} Q(\boldsymbol{\beta}) &= \frac{1}{2}\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \sum_{j=1}^{p} P_\lambda(|\beta_j|) \\ &= \frac{1}{2}\mathbf{y}^T\mathbf{y} - \boldsymbol{\beta}^T\mathbf{X}^T\mathbf{y} + \frac{1}{2}\boldsymbol{\beta}^T\mathbf{X}^T\mathbf{X}\boldsymbol{\beta} + \sum_{j=1}^{p} P_\lambda(|\beta_j|) \end{aligned}$$

By orthogonality $\mathbf{X}^T\mathbf{X}/n = \mathbf{I}$, we have:

$$\begin{aligned} Q(\boldsymbol{\beta}) &= \frac{1}{2}\mathbf{y}^T\mathbf{y} - n\boldsymbol{\beta}^T\mathbf{z} + \frac{n}{2}\boldsymbol{\beta}^T\boldsymbol{\beta} + \sum_{j=1}^{p} P_\lambda(|\beta_j|) \\ &= \sum_{j=1}^{p} \left[\frac{1}{2}(z_j - \beta_j)^2 + P_\lambda(|\beta_j|) \right] + \text{constant} \end{aligned}$$

Therefore, component-wise optimization is possible. □

Thus we only need to discuss:

$$\hat{\beta} = \arg\min_{\beta} \left\{ \frac{1}{2}(z - \beta)^2 + p_\lambda(|\beta|) \right\}$$

where $p_\lambda(|\beta|)$ is given by the specific penalty form.

3.2.2 Closed-form Solutions & Thresholding Functions

Here we compare the solutions (Thresholding Functions) for different penalties.

1. Hard Thresholding (ℓ_0 Penalty)

Penalty: $P_\lambda(\beta) = \lambda I(\beta \neq 0)$
Solution:

$$\hat{\beta}_{\ell_0}(z) = z \cdot \mathbb{I}(|z| > \sqrt{2\lambda})$$

Interpretation: "Keep or Kill". It keeps large coefficients intact but discards small ones entirely. Discontinuous.

2. Soft Thresholding (Lasso / ℓ_1 Penalty)

Penalty: $P_\lambda(\beta) = \lambda|\beta|$
Solution:

$$\hat{\beta}_{\text{Lasso}}(z) = \text{sign}(z)(|z| - \lambda)_+$$

Interpretation: "Shrinkage and Selection". It shifts the estimate towards zero by λ. Continuous but introduces **bias** for large coefficients.

3. Linear Shrinkage (Ridge / ℓ_2 Penalty)

Penalty: $P_\lambda(\beta) = \frac{\lambda}{2}\beta^2$
Solution:

$$\hat{\beta}_{\text{Ridge}}(z) = \frac{z}{1 + \lambda}$$

Interpretation: "Proportional Shrinkage". No selection (solution is never exactly 0).

4. SCAD & MCP (Folded Concave Penalties)

Goal: To bridge the gap between ℓ_0 (unbiased selection) and ℓ_1 (convexity).

- **SCAD Solution**: Acts like Lasso for small z, transitions to Ridge-like, and becomes Hard Thresholding (unbiased) for large $z > \gamma\lambda$.
- **MCP Solution**: Similar to SCAD but with a simpler "Minimax Concave" path.

$$\hat{\beta}_{\text{MCP}}(z) = \begin{cases} 0 & |z| \leq \lambda \\ \frac{\text{sign}(z)(|z|-\lambda)}{1-1/\gamma} & \lambda < |z| \leq \gamma\lambda \\ z & |z| > \gamma\lambda \end{cases}$$

Method	Penalty	Sparsity?	Unbiased?	Convex?
OLS	None	No	Yes	Yes
Subset (ℓ_0)	$\lambda\|\|\beta\|\|_0$	**Yes**	**Yes**	No
Ridge (ℓ_2)	$\lambda\|\|\beta\|\|_2^2$	No	No	**Yes**
Lasso (ℓ_1)	$\lambda\|\|\beta\|\|_1$	**Yes**	No	**Yes**
SCAD / MCP	Non-convex	**Yes**	**Yes**	No

Table 3.1: Trade-offs between different regularization methods. Note that SCAD/MCP achieve both sparsity and unbiasedness for large coefficients, at the cost of convexity.

3.2.3 Summary: Comparison of Thresholding Rules

Lasso Penalty (L_1 Regularization)

Definition 3.6 (Lasso Penalty) The penalty is defined as:

$$P_\lambda^{\text{Lasso}}(\theta) = \lambda|\theta| \tag{3.3}$$

Theorem 3.7 Under orthogonal design, the closed-form solution for Lasso is the soft thresholding function:

$$\hat{\beta}_{\text{Lasso}}(z) = \text{sign}(z)(|z| - \lambda)_+ \tag{3.4}$$

where $(x)_+ = \max(0, x)$.

Proof. Consider the one-dimensional problem:

$$\hat{\beta} = \arg\min_\beta \left\{ \frac{1}{2}(z - \beta)^2 + \lambda|\beta| \right\}$$

The subgradient condition is:

$$-(z - \beta) + \lambda\partial|\beta| \ni 0$$

where $\partial|\beta|$ is the subdifferential of the absolute value function:

$$\partial|\beta| = \begin{cases} \{1\}, & \beta > 0 \\ [-1, 1], & \beta = 0 \\ \{-1\}, & \beta < 0 \end{cases}$$

Case analysis:

- If $z > \lambda$, then $\beta = z - \lambda > 0$
- If $z < -\lambda$, then $\beta = z + \lambda < 0$
- If $|z| \leqslant \lambda$, then $\beta = 0$

Combining these gives the soft thresholding function. □

Ridge Regression Penalty (L_2 Regularization)

Definition 3.8 (Ridge Regression Penalty) Ridge penalty is defined as:

$$P_\lambda^{\text{Ridge}}(\theta) = \frac{\lambda}{2}\theta^2 \tag{3.5}$$

Theorem 3.9 Under orthogonal design, the closed-form solution for ridge regression is:

$$\hat{\beta}_{\text{Ridge}}(z) = \frac{z}{1+\lambda} \tag{3.6}$$

Proof. One-dimensional problem:

$$\hat{\beta} = \arg\min_{\beta} \left\{ \frac{1}{2}(z-\beta)^2 + \frac{\lambda}{2}\beta^2 \right\}$$

Taking derivative:

$$-(z-\beta) + \lambda\beta = 0 \Rightarrow \beta(1+\lambda) = z \Rightarrow \beta = \frac{z}{1+\lambda}$$

□

L_q Regularization ($0 \leqslant q < \infty$)

Definition 3.10 (L_q Penalty) L_q penalty is defined as:

$$P_{\lambda,q}^{\text{Lq}}(\theta) = \lambda|\theta|^q, \quad 0 \leqslant q < \infty \tag{3.7}$$

Theorem 3.11 Under orthogonal design, closed-form solutions for L_q penalty exist only for special q values:

- When $q = 0$, the solution is the hard thresholding function:

$$\hat{\beta}_{\text{L0}}(z) = \begin{cases} z, & \text{if } |z| > \sqrt{2\lambda}, \\ 0, & \text{otherwise.} \end{cases} \tag{3.8}$$

- When $q = 1$, the solution is the soft thresholding function (see Lasso).
- When $q = 2$, the solution is linear shrinkage (see ridge regression).
- For other q values, no closed-form solution exists, requiring numerical methods.

Proof. For $q = 0$, the objective function is:

$$Q(\beta) = \frac{1}{2}(z-\beta)^2 + \lambda I(|\beta| \neq 0)$$

Compare function values at $\beta = 0$ and $\beta = z$:

- When $\beta = 0$, $Q(0) = \frac{1}{2}z^2$

- When $\beta = z$, $Q(z) = \lambda$

Therefore, when $\frac{1}{2}z^2 > \lambda$, choose $\beta = z$, otherwise choose $\beta = 0$.

For $q = 1$ and $q = 2$, proofs are given above.

For general q, the first-order condition is:

$$\beta - z + \lambda q|\beta|^{q-1} \operatorname{sign}(\beta) = 0$$

This is a nonlinear equation with no closed-form solution. □

Remark 3.12 For $0 < q < 1$, the problem is non-convex and may have multiple local minima. For $q > 1$, the problem is convex with a unique solution, but still lacks closed-form expression.

Bridge Regression (L_q Regularization, $q > 0$)

Definition 3.13 (Bridge Regression Penalty) Bridge regression penalty is a special case of L_q penalty ($q > 0$):

$$P^{\text{Bridge}}_{\lambda,q}(\theta) = \lambda|\theta|^q, \quad q > 0 \tag{3.9}$$

Theorem 3.14 Under orthogonal design, the optimal solution for bridge regression satisfies the equation:

$$\beta + \lambda q|\beta|^{q-1} \operatorname{sign}(\beta) = z \tag{3.10}$$

This equation requires numerical solution.

Proof. Objective function:

$$Q(\beta) = \frac{1}{2}(z - \beta)^2 + \lambda|\beta|^q$$

Taking derivative (considering $\beta > 0$):

$$Q'(\beta) = -(z - \beta) + \lambda q\beta^{q-1} = \beta - z + \lambda q\beta^{q-1}$$

Setting derivative to zero gives:

$$\beta + \lambda q\beta^{q-1} = z$$

Similar results hold for $\beta < 0$. □

§ 3.3 LASSO

3.3.1 Basis concepts

Definition 3.15 (LASSO) Given a response vector $\boldsymbol{y} \in \mathbb{R}^n$, a design matrix $\boldsymbol{X} \in \mathbb{R}^{n\times p}$, and a regularization parameter $\lambda > 0$, the LASSO problem is defined as:

$$\min_{\beta\in\mathbb{R}^p} \frac{1}{2n}\|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|_2^2 + \lambda\|\boldsymbol{\beta}\|_1 \tag{3.11}$$

where $\|\cdot\|_2$ denotes the Euclidean norm and $\|\cdot\|_1$ denotes the ℓ_1-norm.

Remark 3.16 The scaling factor $\frac{1}{2n}$ is conventional but not essential. Some formulations use $\frac{1}{2}$ or omit it entirely, which affects the interpretation of λ.

Definition 3.17 (Constrained LASSO Form) The LASSO problem can be equivalently formulated as a constrained optimization problem:

$$\min_{\boldsymbol{\beta} \in \mathbb{R}^p} \frac{1}{2n} \|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|_2^2 \quad \text{subject to} \quad \|\boldsymbol{\beta}\|_1 \leqslant t \tag{3.12}$$

where $t > 0$ is a constraint parameter.

Theorem 3.18 (Lagrangian Duality) For every $\lambda > 0$ in (3.11), there exists a $t > 0$ in (3.12) such that the two problems have the same solution, and vice versa.

Proof. This follows from standard Lagrangian duality theory for convex optimization. The Lagrangian function for (3.12) is:

$$\mathcal{L}(\boldsymbol{\beta}, \mu) = \frac{1}{2n} \|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|_2^2 + \mu(\|\boldsymbol{\beta}\|_1 - t)$$

where $\mu \geq 0$ is the Lagrange multiplier. By strong duality for convex problems, there exists a correspondence between λ and μ. □

3.3.2 Subgradients and Optimality Conditions

Definition 3.19 (Subgradient) Let $f : \mathbb{R}^p \to \mathbb{R}$ be a convex function. A vector $\boldsymbol{g} \in \mathbb{R}^p$ is called a *subgradient* of f at $\boldsymbol{x}$ if for all $\boldsymbol{y} \in \mathbb{R}^p$:

$$f(\boldsymbol{y}) \geqslant f(\boldsymbol{x}) + \boldsymbol{g}^\top (\boldsymbol{y} - \boldsymbol{x})$$

The set of all subgradients at $\boldsymbol{x}$ is called the *subdifferential* and is denoted by $\partial f(\boldsymbol{x})$.

Lemma 3.20 For the absolute value function $f(x) = |x|$, the subdifferential is:

$$\partial |x| = \begin{cases} \{1\} & \text{if } x > 0 \\ \{-1\} & \text{if } x < 0 \\ [-1, 1] & \text{if } x = 0 \end{cases}$$

Theorem 3.21 A vector $\hat{\boldsymbol{\beta}} \in \mathbb{R}^p$ is an optimal solution to the LASSO problem (3.11) if and only if there exists a subgradient vector $\boldsymbol{s} \in \mathbb{R}^p$ with $s_j \in \partial |\hat{\beta}_j|$ for $j = 1, \ldots, p$ such that:

$$-\frac{1}{n} \boldsymbol{X}_j^\top (\boldsymbol{y} - \boldsymbol{X}\hat{\boldsymbol{\beta}}) + \lambda s_j = 0, \quad j = 1, \ldots, p \tag{3.13}$$

where $\boldsymbol{X}_j$ denotes the j-th column of $\boldsymbol{X}$.

Proof. The LASSO objective function is convex, so $\hat{\boldsymbol{\beta}}$ is optimal if and only if $\mathbf{0} \in \partial f(\hat{\boldsymbol{\beta}})$. The subdifferential of the LASSO objective is:

$$\partial \left(\frac{1}{2n} \|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|_2^2 + \lambda \|\boldsymbol{\beta}\|_1 \right) = -\frac{1}{n} \boldsymbol{X}^\top (\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}) + \lambda \partial \|\boldsymbol{\beta}\|_1$$

The result follows by writing this condition coordinate-wise. □

Remark 3.22 The KKT conditions (3.13) can be expressed coordinate-wise as:

$$\text{If } \hat{\beta}_j \neq 0: \quad -\frac{1}{n}\boldsymbol{X}_j^\top(\boldsymbol{y} - \boldsymbol{X}\hat{\boldsymbol{\beta}}) + \lambda \cdot \text{sign}(\hat{\beta}_j) = 0 \tag{3.14}$$

$$\text{If } \hat{\beta}_j = 0: \quad \left|\frac{1}{n}\boldsymbol{X}_j^\top(\boldsymbol{y} - \boldsymbol{X}\hat{\boldsymbol{\beta}})\right| \leqslant \lambda \tag{3.15}$$

Definition 3.23 (Soft Thresholding Operator) The *soft thresholding operator* $S : \mathbb{R}\times\mathbb{R}_+ \to \mathbb{R}$ is defined as:

$$S(z, \lambda) = \text{sign}(z)(|z| - \lambda)_+ = \begin{cases} z - \lambda & \text{if } z > \lambda \\ 0 & \text{if } |z| \leqslant \lambda \\ z + \lambda & \text{if } z < -\lambda \end{cases} \tag{3.16}$$

where $(x)_+ = \max(0, x)$.

Orthogonal Design Case

Theorem 3.24 If the columns of $\boldsymbol{X}$ are orthogonal, i.e., $\boldsymbol{X}^\top\boldsymbol{X} = n\boldsymbol{I}_p$, then the LASSO solution has the explicit form:

$$\hat{\beta}_j = S\left(\frac{1}{n}\boldsymbol{X}_j^\top\boldsymbol{y}, \lambda\right), \quad j = 1, \ldots, p \tag{3.17}$$

Proof. Under orthogonal design $\boldsymbol{X}^\top\boldsymbol{X} = n\boldsymbol{I}_p$, the OLS estimator is $\hat{\boldsymbol{\beta}}^{\text{OLS}} = \frac{1}{n}\boldsymbol{X}^\top\boldsymbol{y}$. The KKT conditions become:

$$-(\hat{\beta}_j^{\text{OLS}} - \hat{\beta}_j) + \lambda s_j = 0, \quad s_j \in \partial|\hat{\beta}_j|$$

Consider three cases:

- If $\hat{\beta}_j > 0$, then $s_j = 1$ and $\hat{\beta}_j = \hat{\beta}_j^{\text{OLS}} - \lambda$
- If $\hat{\beta}_j < 0$, then $s_j = -1$ and $\hat{\beta}_j = \hat{\beta}_j^{\text{OLS}} + \lambda$
- If $\hat{\beta}_j = 0$, then $s_j \in [-1, 1]$ and $|\hat{\beta}_j^{\text{OLS}}| \leqslant \lambda$

This exactly matches the soft thresholding operator definition (3.16). □

Proposition 3.25 (Shrinkage) The LASSO estimator shrinks coefficients toward zero:

$$|\hat{\beta}_j| \leqslant |\hat{\beta}_j^{\text{OLS}}|, \quad j = 1, \ldots, p$$

with equality only when $\hat{\beta}_j^{\text{OLS}} = 0$ or $\lambda = 0$.

3.3.3 Algorithms

Coordinate Descent for LASSO

Coordinate descent is an iterative optimization algorithm that minimizes a multivariate function by sequentially optimizing along coordinate directions. For the LASSO problem, this approach updates one coefficient β_j at a time while keeping all other coefficients fixed, reducing the complex multivariate optimization to a sequence of simple univariate problems.

Theorem 3.26 When optimizing coordinate j with all other coefficients β_k ($k \neq j$) fixed, the LASSO subproblem reduces to:

$$\min_{\beta_j \in \mathbb{R}} \left\{ \frac{1}{2n} \sum_{i=1}^{n} (r_i^{(j)} - x_{ij}\beta_j)^2 + \lambda|\beta_j| \right\}$$

where $r_i^{(j)} = y_i - \sum_{k \neq j} x_{ik}\beta_k$ are the partial residuals. The optimal solution is given by the soft-thresholding operator (3.16):

$$\hat{\beta}_j = \frac{S(\rho_j, \lambda)}{z_j} \tag{3.18}$$

where:

$$\rho_j = \frac{1}{n}\langle \boldsymbol{x}_j, \boldsymbol{r}^{(j)} \rangle = \frac{1}{n}\sum_{i=1}^{n} x_{ij} r_i^{(j)}$$

$$z_j = \frac{1}{n}\|\boldsymbol{x}_j\|_2^2 = \frac{1}{n}\sum_{i=1}^{n} x_{ij}^2$$

Proof. The subgradient optimality condition for the coordinate-wise problem is:

$$-\frac{1}{n}\sum_{i=1}^{n} x_{ij}(r_i^{(j)} - x_{ij}\beta_j) + \lambda s_j = 0, \quad s_j \in \partial|\beta_j|$$

Rewriting this equation:

$$-\frac{1}{n}\sum_{i=1}^{n} x_{ij} r_i^{(j)} + \left(\frac{1}{n}\sum_{i=1}^{n} x_{ij}^2 \right)\beta_j + \lambda s_j = 0$$

Substituting the definitions:

$$-\rho_j + z_j\beta_j + \lambda s_j = 0$$

This corresponds exactly to the optimality condition for the univariate soft-thresholding problem, yielding the solution in equation (3.18). □

Theorem 3.27 The coordinate-wise update can be equivalently expressed using the full residual vector $\boldsymbol{r} = \boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}$:

$$\hat{\beta}_j \leftarrow \frac{1}{z_j} S\left(\hat{\beta}_j z_j + \frac{1}{n}\langle \boldsymbol{x}_j, \boldsymbol{r} \rangle, \lambda \right) \tag{3.19}$$

This formulation avoids explicit computation of partial residuals and enables efficient residual maintenance.

Proof. Observe the relationship between partial and full residuals:

$$r_i^{(j)} = y_i - \sum_{k \neq j} x_{ik}\beta_k = \left(y_i - \sum_{k=1}^{p} x_{ik}\beta_k\right) + x_{ij}\beta_j = r_i + x_{ij}\beta_j$$

Substitute into the expression for ρ_j:

$$\begin{aligned}
\rho_j &= \frac{1}{n}\sum_{i=1}^{n} x_{ij} r_i^{(j)} \\
&= \frac{1}{n}\sum_{i=1}^{n} x_{ij}(r_i + x_{ij}\beta_j) \\
&= \frac{1}{n}\sum_{i=1}^{n} x_{ij} r_i + \beta_j \cdot \frac{1}{n}\sum_{i=1}^{n} x_{ij}^2 \\
&= \frac{1}{n}\langle \boldsymbol{x}_j, \boldsymbol{r}\rangle + \beta_j z_j
\end{aligned}$$

Substituting this into equation (3.18) proves the equivalent update formula. □

Algorithm 5 Coordinate Descent for LASSO

Input: Design matrix $\boldsymbol{X} \in \mathbb{R}^{n\times p}$, response vector $\boldsymbol{y} \in \mathbb{R}^n$, regularization parameter $\lambda \geqslant 0$, tolerance $\epsilon > 0$

Output: LASSO solution $\hat{\boldsymbol{\beta}}$

```
Initialize β ← 0 ∈ ℝ^p
Precompute column norms: z_j ← ||x_j||_2^2 / n for j = 1, ..., p
Compute initial residual: r ← y − Xβ
repeat
    β_old ← β                                          ▷ Store previous coefficients
    for j = 1 to p do                                  ▷ Cycle through coordinates
        Compute correlation: α_j ← ⟨x_j, r⟩/n
        Compute soft-threshold input: ρ̃_j ← β_j z_j + α_j
        Compute coordinate update: β_j^new ← S(ρ̃_j, λ)/z_j
        Update residual: r ← r + x_j(β_j − β_j^new)
        Update coefficient: β_j ← β_j^new
    end for
until ||β − β_old||_2 < ε                              ▷ Check convergence
β̂ ← β
```

Least Angle Regression

Least Angle Regression (LARS) is an efficient algorithm for computing the entire regularization path of LASSO solutions. Unlike forward stepwise regression which adds variables in discrete jumps, LARS proceeds in a piecewise linear fashion, adding variables to the active set in a way that maintains equal absolute correlation with the current residual.

Algorithm 6 Least Angle Regression (LARS) for LASSO

Input: Design matrix $\boldsymbol{X} \in \mathbb{R}^{n \times p}$, response vector $\boldsymbol{y} \in \mathbb{R}^n$
Output: Complete LASSO solution path $\{\hat{\boldsymbol{\beta}}(\lambda) : \lambda \geq 0\}$

Initialize: $\boldsymbol{\mu} \leftarrow \mathbf{0}$, active set $\mathcal{A} \leftarrow \varnothing$, residual $\boldsymbol{r} \leftarrow \boldsymbol{y} - \boldsymbol{\mu}$
Compute correlations: $\boldsymbol{c} \leftarrow \boldsymbol{X}^\top \boldsymbol{r}$
while $\max_j |c_j| > 0$ **do**
    $j^* \leftarrow \arg\max_j |c_j|$
    $\mathcal{A} \leftarrow \mathcal{A} \cup \{j^*\}$
    Compute equiangular direction $\boldsymbol{u}_{\mathcal{A}}$ and step size γ
    repeat
        $\boldsymbol{\mu} \leftarrow \boldsymbol{\mu} + \gamma \boldsymbol{u}_{\mathcal{A}}$
        $\boldsymbol{r} \leftarrow \boldsymbol{y} - \boldsymbol{\mu}$
        $\boldsymbol{c} \leftarrow \boldsymbol{X}^\top \boldsymbol{r}$
        if any β_j in $\mathcal{A}$ changes sign **then**
            Remove that j from $\mathcal{A}$
            break
        end if
    until active set changes or $\max_j |c_j| = 0$
    Store solution $\hat{\boldsymbol{\beta}}(\lambda)$ with $\lambda = \max_j |c_j|/n$
end while

3.3.4 Bayesian Interpretation

Theorem 3.28 The LASSO estimator is equivalent to the maximum a posteriori (MAP) estimator under independent Laplace priors:

$$\beta_j \sim \text{Laplace}(0, 1/\lambda), \quad j = 1, \dots, p$$

Proof. The Laplace prior density is $p(\beta_j) = \frac{\lambda}{2} e^{-\lambda|\beta_j|}$. The posterior distribution is:

$$p(\boldsymbol{\beta}|\boldsymbol{y}) \propto \exp\left(-\frac{1}{2\sigma^2}\|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|_2^2\right) \prod_{j=1}^{p} e^{-\lambda|\beta_j|}$$

Maximizing the log-posterior gives the LASSO objective (3.11). □

§ 3.4

Variable Selection Consistency and Oracle Properties

3.4.1 Fundamental Properties of Penalty Functions

Definition 3.29 (Unbiasedness) A penalty function $p_\lambda(\cdot)$ satisfies the *unbiasedness* property if:

$$\lim_{|\beta|\to\infty} p'_\lambda(|\beta|) = 0$$

This ensures that large coefficients are not excessively shrunk:

$$\lim_{|\beta_j|\to\infty} \left|\mathbb{E}[\hat{\beta}_j] - \beta_j\right| = 0$$

Definition 3.30 (Sparsity) A penalty function $p_\lambda(\cdot)$ induces *sparsity* if:

$$p'_\lambda(0+) = \lim_{t\to 0^+} \frac{p_\lambda(t)}{t} > 0$$

This guarantees the existence of a threshold $\tau > 0$ such that:

$$|\hat{\beta}_j| < \tau \Rightarrow \hat{\beta}_j = 0$$

Proposition 3.31 (Sparsity Condition) From the KKT optimality conditions, sparsity requires:

$$\hat{\beta}_j = 0 \quad \text{if and only if} \quad \left|\frac{1}{n}\boldsymbol{X}_j^\top(\boldsymbol{y} - \boldsymbol{X}\hat{\boldsymbol{\beta}}_{-j})\right| \leq \lambda p'_\lambda(0+)$$

Verification for Common Penalties. All common penalties satisfy $p'_\lambda(0+) = \lambda > 0$:

- **LASSO**: $\lambda > 0 \Rightarrow$ **Sparse**
- **SCAD**: $\lambda > 0 \Rightarrow$ **Sparse**
- **MCP**: $\lambda > 0 \Rightarrow$ **Sparse**
- ℓ_0: $p'_\lambda(0+) = \infty \Rightarrow$ **Sparse**

□

Definition 3.32 (Continuity) A penalty function produces *continuous* estimators if:

$$\sup_{t>0} \left|\frac{p'_\lambda(t)}{t}\right| < \infty$$

This ensures the solution path $\lambda \mapsto \hat{\boldsymbol{\beta}}(\lambda)$ is continuous:

$$\forall\epsilon > 0, \exists\delta > 0 : \|\boldsymbol{y} - \boldsymbol{y}'\| < \delta \Rightarrow \|\hat{\boldsymbol{\beta}}(\boldsymbol{y}) - \hat{\boldsymbol{\beta}}(\boldsymbol{y}')\| < \epsilon$$

Proposition 3.33 (Continuity Conditions) Continuity is guaranteed when:

1. The optimization problem is convex, or
2. The penalty function has bounded second derivative, or
3. The penalty satisfies the Lipschitz condition

Verification for Common Penalties.

- **LASSO**: Convex $\Rightarrow$ **Continuous**
- **SCAD**: Piecewise continuous derivative $\Rightarrow$ **Continuous**
- **MCP**: Continuous derivative $\Rightarrow$ **Continuous**
- ℓ_0: Discontinuous $\Rightarrow$ **Not Continuous**

□

3.4.2 Unified Mathematical Framework

General Optimization Problem

Consider the regularized estimation:

$$\min_{\beta \in \mathbb{R}^p} \frac{1}{2n} \|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|_2^2 + \lambda \sum_{j=1}^{p} p(|\beta_j|)$$

The Karush-Kuhn-Tucker (KKT) optimality conditions are:

$$\frac{1}{n} \boldsymbol{X}_j^\top (\boldsymbol{y} - \boldsymbol{X}\hat{\boldsymbol{\beta}}) = \begin{cases} \lambda p'_\lambda(|\hat{\beta}_j|) \cdot \operatorname{sign}(\hat{\beta}_j) & \text{if } \hat{\beta}_j \neq 0 \\ \lambda p'_\lambda(0+) \cdot u_j & \text{if } \hat{\beta}_j = 0 \end{cases}$$

where $|u_j| \leq 1$.

Property Interrelationships

Theorem 3.34 (Property Trade-offs) The three properties exhibit fundamental trade-offs:

1. **Unbiasedness + Sparsity** $\Rightarrow$ Non-convexity
2. **Convexity** $\Rightarrow$ Either biased or non-sparse
3. **Continuity** is incompatible with hard thresholding

Proof.

- If $p'_\lambda(0+) > 0$ (sparsity) and $p'_\lambda(|\beta|) \to 0$ (unbiasedness), then p_λ must be non-linear
- Linear penalties (convex) cannot satisfy both endpoint conditions
- Hard thresholding (like ℓ_0) is inherently discontinuous

□

3.4.3 Oracle Properties Framework

Definition 3.35 (Oracle Properties) A parameter estimator $\hat{\boldsymbol{\beta}}$ is said to have the *oracle properties* if it satisfies:

1. **Variable Selection Consistency**:

$$\lim_{n\to\infty} P(\{j : \hat{\beta}_j \neq 0\} = \mathcal{A}^*) = 1$$

   where $\mathcal{A}^* = \{j : \beta_j^* \neq 0\}$ is the true active set.

2. **Asymptotic Normality**: For the non-zero coefficients $\hat{\boldsymbol{\beta}}_{\mathcal{A}^*}$, we have

$$\sqrt{n}(\hat{\boldsymbol{\beta}}_{\mathcal{A}^*} - \boldsymbol{\beta}^*_{\mathcal{A}^*}) \xrightarrow{d} N(\mathbf{0}, \Sigma^*)$$

   where Σ^* is the covariance matrix of the oracle estimator that knows $\mathcal{A}^*$ in advance.

Theorem 3.36 (Oracle Property Requirements) For a penalty function to achieve oracle properties, it must satisfy:

1. **Unbiasedness**: $\lim_{|\beta|\to\infty} p'_\lambda(|\beta|) = 0$
2. **Sparsity**: $p'_\lambda(0+) > 0$
3. **Regularity**: $\lambda_n \to 0$ and $\sqrt{n}\lambda_n \to \infty$
4. **Continuity**: $\sup_{t>0} \left|\frac{p'_\lambda(t)}{t}\right| < \infty$

3.4.4 Advanced Penalty Functions

SCAD Penalty

Definition 3.37 (Smoothly Clipped Absolute Deviation (SCAD)) The SCAD penalty is defined through its derivative:

$$p'_\lambda(|\beta|) = \lambda \left\{ I(|\beta| \leq \lambda) + \frac{(a\lambda - |\beta|)_+}{(a-1)\lambda} I(|\beta| > \lambda) \right\}$$

for some $a > 2$, where $I(\cdot)$ is the indicator function.

Proposition 3.38 (SCAD Penalty Function) The SCAD penalty function has the explicit form:

$$p_\lambda(|\beta|) = \begin{cases} \lambda|\beta| & \text{if } |\beta| \leq \lambda \\ \frac{2a\lambda|\beta| - \beta^2 - \lambda^2}{2(a-1)} & \text{if } \lambda < |\beta| \leq a\lambda \\ \frac{\lambda^2(a+1)}{2} & \text{if } |\beta| > a\lambda \end{cases}$$

Theorem 3.39 (Properties of SCAD) The SCAD penalty possesses the following key properties:

1. **Unbiasedness**: For large $|\beta|$, $p'_\lambda(|\beta|) = 0$, eliminating bias
2. **Sparsity**: $p'_\lambda(0+) = \lambda$, ensuring sparse solutions
3. **Continuity**: The resulting estimator is continuous in λ

Property Verification for SCAD.

$$\begin{aligned} \textbf{Unbiasedness:} &\quad \lim_{|\beta|\to\infty} p'_\lambda(|\beta|) = 0 \\ \textbf{Sparsity:} &\quad p'_\lambda(0+) = \lambda > 0 \\ \textbf{Continuity:} &\quad \sup_{t>0} \left|\frac{p'_\lambda(t)}{t}\right| = \frac{1}{a-1} < \infty \end{aligned}$$

□

Theorem 3.40 (Oracle Properties of SCAD) Under regularity conditions:

- $\frac{1}{n}\boldsymbol{X}^\top\boldsymbol{X} \to \boldsymbol{C}$ (positive definite)
- $\lambda_n \to 0$ and $\sqrt{n}\lambda_n \to \infty$
- The true model is sparse

the SCAD estimator possesses the oracle properties.

MCP Penalty

Definition 3.41 (Minimax Concave Penalty (MCP)) The MCP penalty is defined as:

$$p_\lambda(|\beta|) = \begin{cases} \lambda|\beta| - \frac{\beta^2}{2a} & \text{if } |\beta| \le a\lambda \\ \frac{1}{2}a\lambda^2 & \text{if } |\beta| > a\lambda \end{cases}$$

for some $a > 1$.

Proposition 3.42 (MCP Derivative) The derivative of MCP is:

$$p'_\lambda(|\beta|) = \begin{cases} \lambda - \frac{|\beta|}{a} & \text{if } |\beta| \le a\lambda \\ 0 & \text{if } |\beta| > a\lambda \end{cases}$$

Theorem 3.43 (Properties of MCP) MCP shares the oracle properties with SCAD but has additional advantages:

1. **Minimax Optimality**: MCP minimizes the maximum concavity
2. **Simple Thresholding**: The solution has closed-form thresholding representation
3. **Computational Efficiency**: Easier to optimize than SCAD

Property Verification for MCP.

$$\begin{aligned} \textbf{Unbiasedness:} \quad & \lim_{|\beta|\to\infty} p'_\lambda(|\beta|) = 0 \\ \textbf{Sparsity:} \quad & p'_\lambda(0+) = \lambda > 0 \\ \textbf{Continuity:} \quad & \sup_{t>0} \left| \frac{p'_\lambda(t)}{t} \right| = \frac{1}{a} < \infty \end{aligned}$$

□

Theorem 3.44 (MCP Thresholding Rule) The MCP estimator has the explicit thresholding representation:

$$\hat{\beta}_j = \begin{cases} \frac{S(\frac{1}{n}\boldsymbol{X}_j^\top \boldsymbol{y}, \lambda)}{1-1/a} & \text{if } |\frac{1}{n}\boldsymbol{X}_j^\top \boldsymbol{y}| \le a\lambda \\ \frac{1}{n}\boldsymbol{X}_j^\top \boldsymbol{y} & \text{if } |\frac{1}{n}\boldsymbol{X}_j^\top \boldsymbol{y}| > a\lambda \end{cases}$$

where $S(\cdot, \lambda)$ is the soft-thresholding operator.